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(54) **NUCLEIC ACID THERAPEUTICS FOR GENETIC DISORDERS**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Provided herein, are compositions based on retroviruses
(e.g., lentiviruses) comprising one or more nucleic acid
molecules encoding retroviral Pol polyprotein components
and a nucleic acid molecule comprising one or more trans-
gene sequences flanked by long terminal repeat sequences,
for delivery of the one or more transgenes to a target cell ex
vivo or in vivo. The compositions are useful for delivering
to a target cell (e.g., hematopoietic stem cells (HSCs), liver
cells, ocular cells, muscle cells, epithelial cells, T cells, etc.)
and/or stably expressing any transgene (e.g., beta-globin,
Factor VIII, RP GTPase regulator (RPGR), dystrophin,
cystic fibrosis transmembrane conductance regulator
(CFTR), a chimeric antigen receptor, etc.) with a biological
effect to treat and/or ameliorate the symptoms associated
with any disorder related to gene expression (e.g., sickle cell
disease, beta-thalassemia, haemophilia B, retinitis pigmen-
tosa, Duchenne muscular dystrophy, cystic fibrosis, cancer,
etc.).

24 Claims, 11 Drawing Sheets

Specification includes a Sequence Listing.

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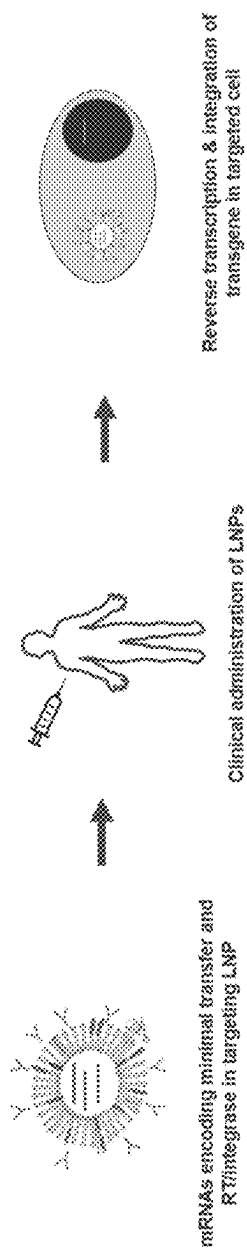


FIG. 1

	Non-dividing cells ^a	Genome integration	Expression site	Transgene capacity	Production titer	Major advantage	Major challenge
CRISPR/Cas9		✓	Site-specific	++	NA	Gene editing	Edits dividing cells only
Retrovirus		✓	Actively transcribed genes	++	+	Integration	Edits dividing cells only; potential genotoxicity
Lentivirus	✓	✓	Actively transcribed genes	++	++	Integration	Potential genotoxicity
Herpesvirus	✓		Episomal	+++++	+++	Transgene capacity	Dosing requirements
Adenovirus	✓		Episomal	+++	+++	Production titer	Dosing requirements
AAV	✓		Episomal	+	+++++	Production titer	Capacity, potency, repeated dosing

FIG. 2



FIG. 3

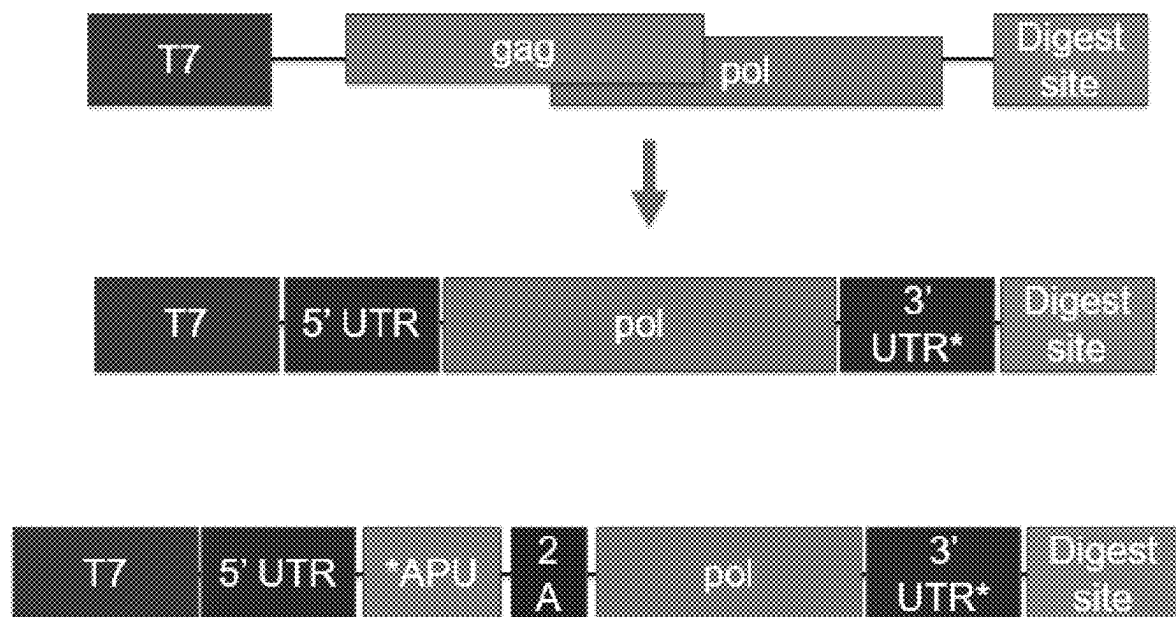


FIG. 4

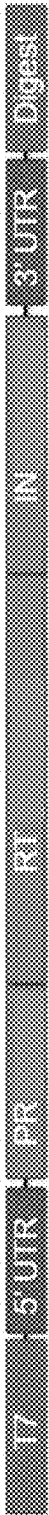


FIG. 5A

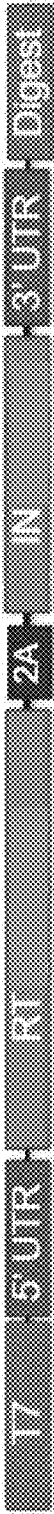


FIG. 5B

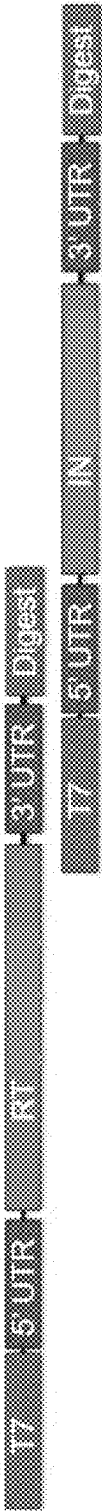


FIG. 5C

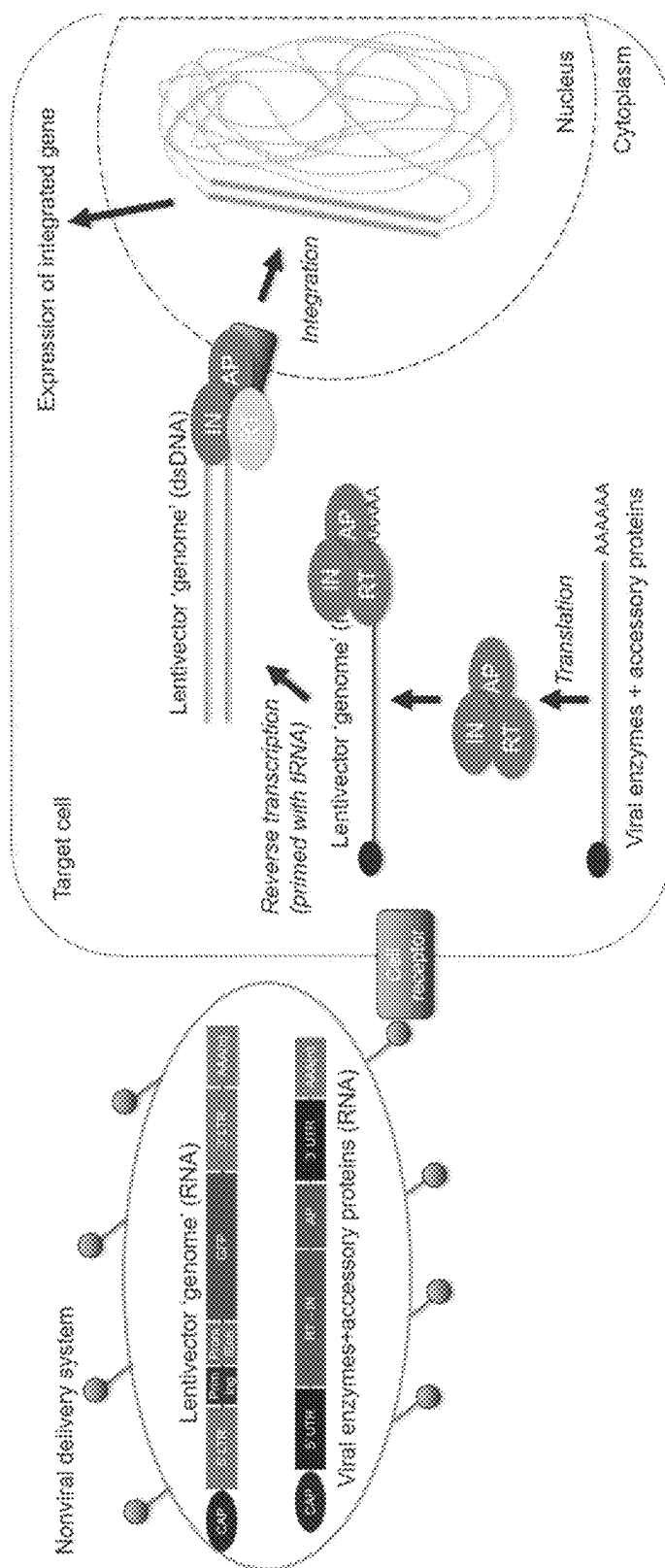


FIG. 6

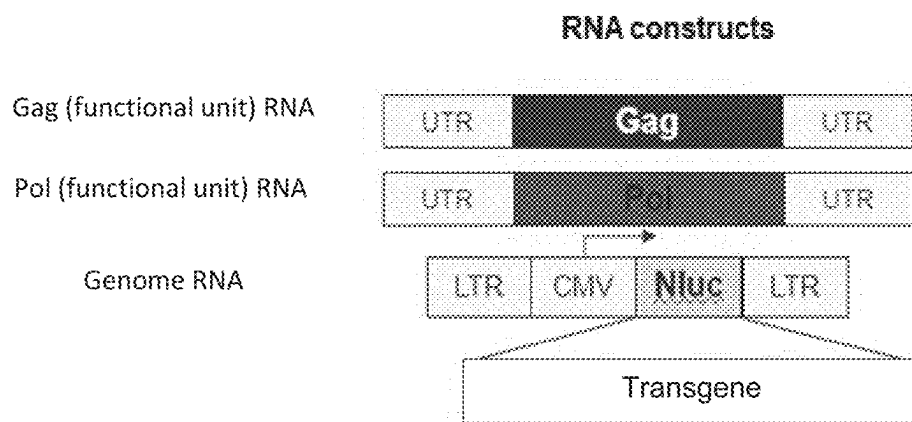


FIG. 7A

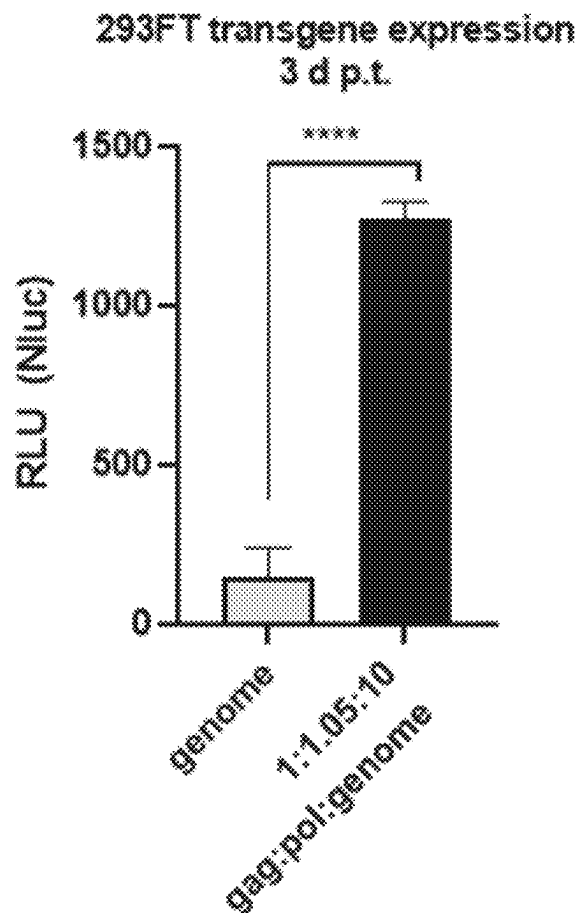


FIG. 7B

Integrase Functional Unit RNA Constructs

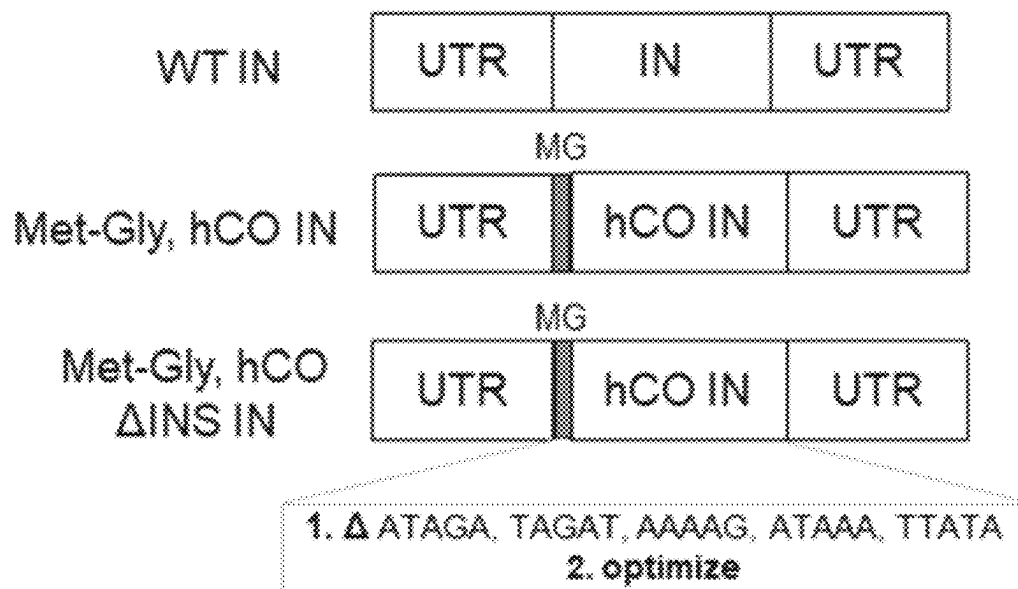


FIG. 8A

Met-Gly, hCO Δ INS IN RNA - Poly(A) +Poly(A)*

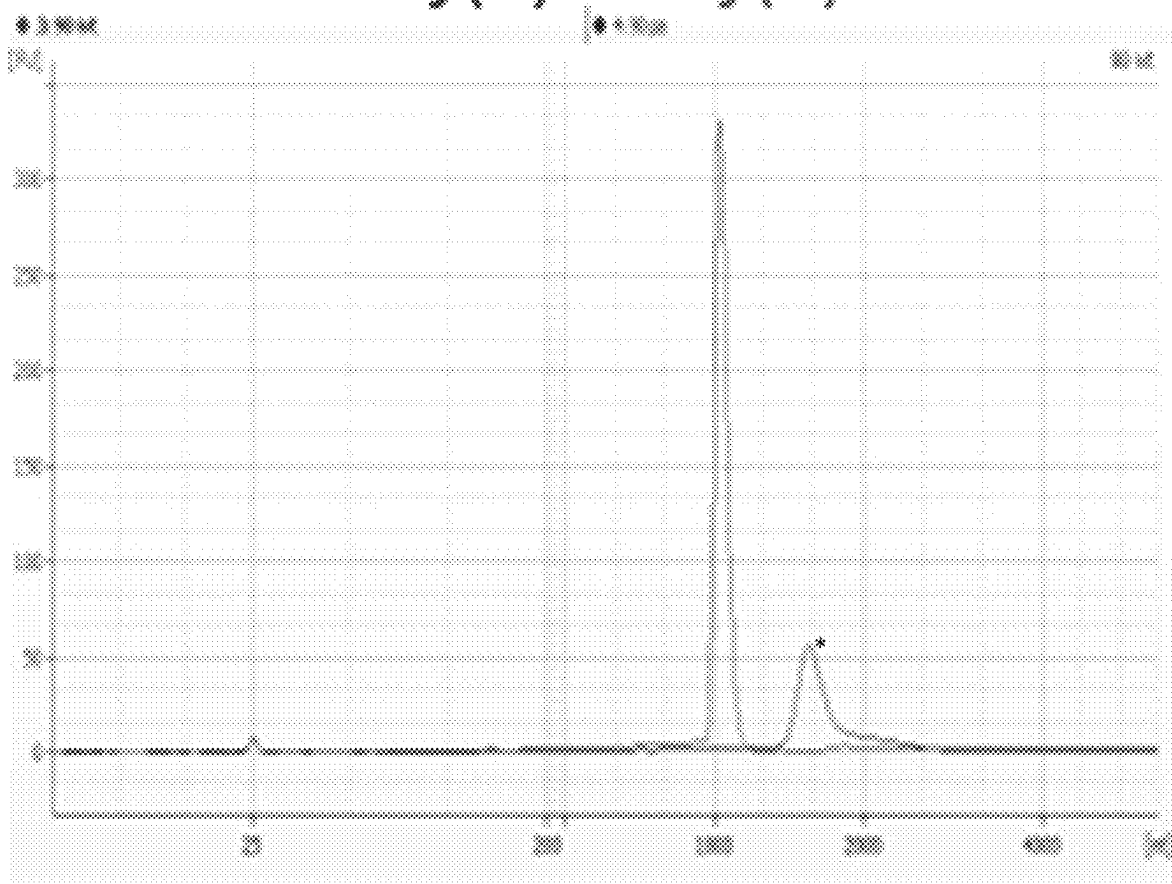


FIG. 8B

Integrase Expression, 48 h p.t.

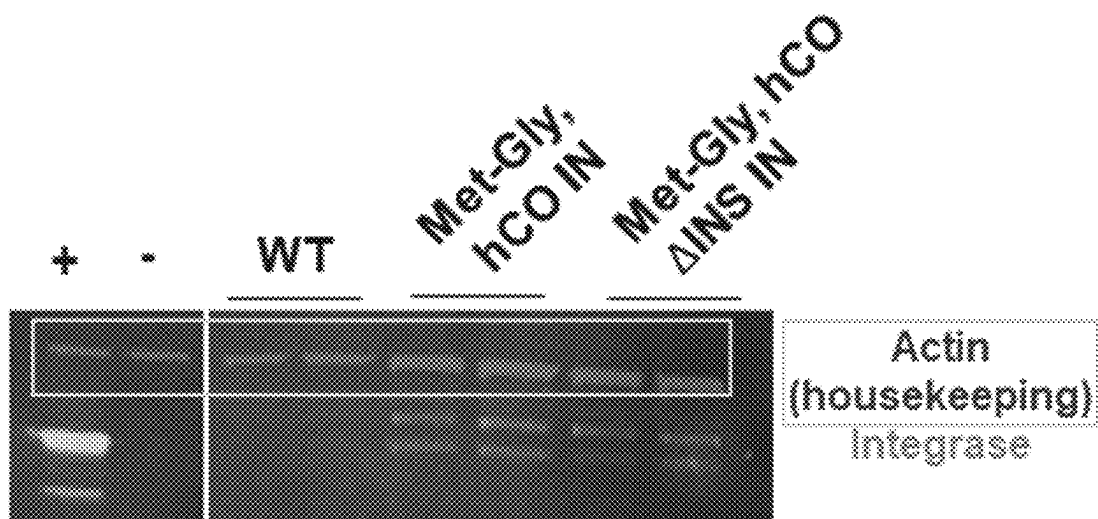


FIG. 8C

Subcellular Integrase Localization, Met-Gly, hCO ΔINS IN RNA, 24 h p.t.

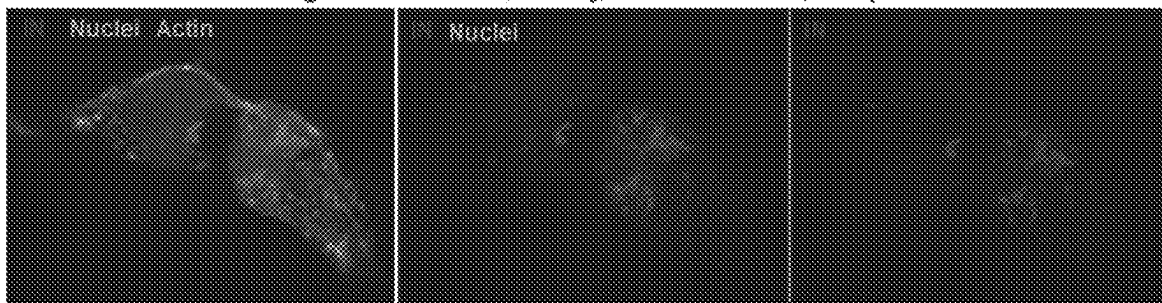


FIG. 8D

HBG UTR	p51	HBG UTR
HBG UTR	p66	HBG UTR

FIG. 9

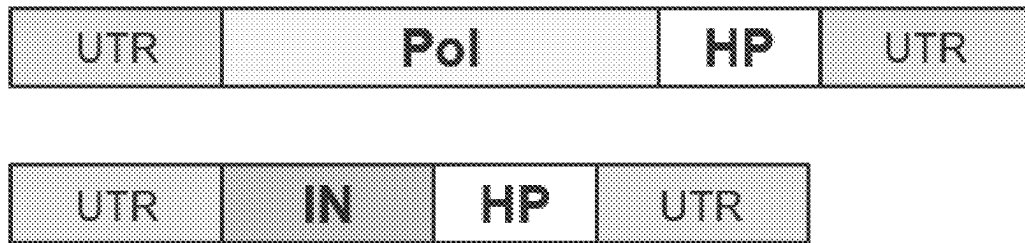


FIG. 10

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NUCLEIC ACID THERAPEUTICS FOR GENETIC DISORDERS

RELATED APPLICATION

This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/053,474, filed Jul. 17, 2020, the content of which is incorporated herein by reference in its entirety.

REFERENCE TO THE SEQUENCE LISTING

The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled G0830.70039US01-SEQ.txt created on Jul. 16, 2021, which is 75,239 bytes in size. The information in electronic format of the Sequence Listing is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Examples of current gene therapy approaches include, but are not limited to, CRISPR-based systems, such as CRISPR/Cas9, retroviruses, lentiviruses, herpesviruses, adenoviruses, and adeno-associated viruses (AAV), as seen in FIG. 2. Although there are many advantages to these various approaches, significant limitations also exist. For example, retroviruses and some CRISPR-based systems generally require cell division and homologous recombination in order to be successful, and therefore, do not function in non-dividing cells. In AAVs, there exists a transgene size limitation, effectively limiting their capacity and usefulness. Other limitations of adenoviruses, include, for example, toxicity associated with high doses due to limited efficacy and a decrease in the effectiveness of repeat dosing due to the immune response to the virus. Moreover, the production of the virus for a therapeutic dose can be quite costly.

As shown in FIG. 2, lentiviruses have previously been used as a model for gene therapy programs. Lentiviruses are ssRNA genome viruses (e.g., HIV) that infect a wide range of hosts and integrate into the genome, sometimes in a latent DNA form. In current gene therapy programs, lentiviral particles are produced using a safer, split construct design. For example, the third-generation design involves gene therapy using four plasmids: a transfer plasmid which carries the gene of interest and integrates into the genome, a packaging plasmid that provides the particle formation, reverse transcription, and integration functions, the envelope plasmid which provides a broad host range, and a plasmid that provides the nuclear export of mRNA. A general limitation of the lentiviral approach is that the production of the virus requires expansion in cells, which is slow and expensive; and the processing of viral particles is inefficient.

Thus, there remains a need for improved gene delivery systems beyond those available today.

SUMMARY OF THE INVENTION

Disclosed herein are nucleic acid compositions for in vivo gene therapy. The compositions include a minimal design which provide transgene-encoding nucleic acid molecules, nucleic acid molecules expressing reverse transcriptase and/or integrase, and can contain auxiliary nucleic acid molecules that support reverse transcription and integration of the transgenes, and can contain a delivery system. The compositions do not require nucleic acid sequences that express proteins encoded by the retroviral rev and env genes.

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However, the compositions can contain nucleic acid sequences that express one or more proteins encoded by the retroviral rev gene, the retroviral env gene, or both. The invention also provides nucleic acid molecules, methods of use, methods of treatment, nucleic acid templates, cells, and kits. The invention also provides for modifications to increase the stability and/or functioning of the nucleic acid and/or the stability and/or functioning of the encoded protein. These modifications can include mutations in one or more intrinsic instability (INS) elements, a functional unit, such as an integrase, with an N-terminal methionine-glycine dipeptide; fusion of a functional unit, such as an integrase, to a homing protein; use of codon optimization for expression in a host cell; and/or use of a priming oligonucleotide. Other modifications are as disclosed herein.

Some aspects of the invention include one or more nucleic acid molecules encoding one or more Pol polyprotein components flanked by 5' and 3' untranslated regions (UTRs), and a nucleic acid molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR. Some aspects of the invention include one or more nucleic acid molecules encoding one or more of the protein components of the Pol polyprotein flanked by 5' and 3' untranslated regions (UTRs), and a nucleic acid molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, but do not include nucleic acid sequences that express proteins encoded by at least one of retroviral rev gene and retroviral env gene. Some aspects of the invention include one or more nucleic acid molecules encoding one or more Pol polyprotein components flanked by 5' and 3' untranslated regions (UTRs), and a nucleic acid molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, but do not include nucleic acid sequences that express proteins encoded by both the retroviral rev and env genes. Some aspects of the invention include one or more nucleic acid molecules encoding one or more Pol polyprotein components flanked by 5' and 3' untranslated regions (UTRs), and a nucleic acid molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, and are capable of integrating the one or more transgenes into a host genome in the absence of functional retroviral Rev and/or Env proteins. The expression of the Pol polyprotein components does not necessarily require translational slippage from an inline gag gene. The one or more nucleic acid molecules encoding the Pol polyprotein components may be a nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding a Pol polyprotein, and a 3' UTR. Alternatively, the one or more nucleic acid molecules encoding the Pol polyprotein components may be a nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein components reverse transcriptase and integrase, and a 3' UTR. The Pol polyprotein components reverse transcriptase and integrase may be expressed on a polycistronic construct. In some embodiments, the polycistronic construct is bicistronic. In some embodiments, the polycistronic construct is tricistronic. The Pol polyprotein components reverse transcriptase and integrase may be expressed with one or more polycistronic elements. The polycistronic element may be an

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intervening internal ribosome entry site (IRES), a2A peptide-encoding sequence, or other polycistronic element. The one or more nucleic acid molecules encoding the Pol polyprotein components may be two nucleic acid molecules, wherein one of them comprises a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein component reverse transcriptase, and a 3' UTR, and the other one comprises a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein component integrase, and a 3' UTR. FIGS. 3-5C illustrate exemplary embodiments of some aspects of the invention.

Some aspects of the invention provide a composition that is packaged in a non-viral delivery system, wherein the composition comprises a first RNA molecule that includes a 5' untranslated region (UTR), a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR; and a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR. Some aspects of the invention provide a composition that is packaged in a non-viral delivery system, wherein the composition comprises a first RNA molecule that includes a 5' untranslated region (UTR), a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR; and a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, but does not include nucleic acid sequences that express proteins encoded by at least one of retroviral rev gene and retroviral env gene. Some aspects of the invention provide a composition that is packaged in a non-viral delivery system, wherein the composition comprises a first RNA molecule that includes a 5' untranslated region (UTR), a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR; and a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, but does not include nucleic acid sequences that express proteins encoded by both the retroviral rev and env genes. Some aspects of the invention provide a composition that is packaged in a non-viral delivery system, wherein the composition comprises a first RNA molecule that includes a 5' untranslated region (UTR), a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR and a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR, and that is capable of integrating the one or more transgenes into a host genome in the absence of functional retroviral Rev and/or Env proteins.

In some aspects, the one or more nucleic acid molecules encoding the Pol polyprotein components comprise nucleic acid sequences encoding one or more functional units. In some aspects, the one or more nucleic acid molecules encoding the Pol polyprotein components may also comprise nucleic acid sequences encoding one or more accessory proteins. The one or more accessory protein may be selected from nucleocapsid (NC), capsid protein (CA), matrix protein (MA), p6, viral infectivity factor (Vif), transactivator-of transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu). In some aspects, the accessory protein may contain one or more mutations. In some aspects, the CA is a mutant CA. In some aspects, the mutant CA has a N74D and/or E45A mutation,

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with the numbering corresponding to the wild-type HIV-1 capsid protein. In some aspects, the wild-type HIV-1 strain can be NL4-3. In some aspects, the one or more nucleic acid molecules encoding the Pol polyprotein components may also comprise nucleic acid sequences encoding one or more Gag polyprotein accessory proteins (i.e., an accessory protein that is a Gag polyprotein component). The one or more accessory proteins may be encoded on the same nucleic acid molecule as the Pol polyprotein components. The one or more accessory proteins may also be expressed with one or more polycistronic elements. The polycistronic element may be an intervening IRES, a 2A peptide-encoding sequence, or other polycistronic element. Alternatively, the one or more accessory proteins may be encoded by one or more nucleic acid molecules different from the nucleic acid molecule encoding the Pol polyprotein components, wherein each nucleic acid molecule comprises a 5' UTR and a 3' UTR. In some embodiments, the Gag polyprotein accessory proteins may be selected from NC, CA, MA, and p6. In some embodiments, the Gag polyprotein accessory proteins may be encoded by the Gag polyprotein. In other embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components do not encode the Gag polyprotein. In some embodiments, the accessory protein MA is not encoded by any of the nucleic acid molecules. In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components comprises a gag-pol gene. In some embodiments, the gag-pol gene comprises a frameshift mutation. In some embodiments, the frameshift mutation is a single nucleotide insertion or deletion.

Some aspects of the invention provide that the one or more nucleic acid molecules encoding the Pol polyprotein components and/or the accessory proteins are modified to increase the stability of the nucleic acid molecules, and/or to increase the stability of the encoded proteins. In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or the accessory proteins are RNA molecules (e.g., mRNA molecules) and can optionally comprise one or more modifications to increase the stability of the RNA molecules (e.g., mRNA molecules). In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or the accessory proteins are DNA molecules and can optionally comprise one or more modifications to increase the stability of the DNA molecules and/or the corresponding mRNA molecules. In some embodiments, the nucleic acid molecules encoding the one or more Pol polyprotein components and/or the accessory proteins comprise one or more mutations in one or more intrinsic instability (INS) elements. The one or more INS elements may be selected from TAGAT, ATAGA, AAAAG, ATAAA, TTATA, or the like. In some embodiments, a codon in an INS element is mutated to an alternate codon for the same amino acid such that the functional protein sequence is preserved (i.e., a silent mutation) while the INS element is altered or removed. In some embodiments, the one or more nucleic acid molecules encoding the one or more Pol polyprotein components encode an integrase polypeptide comprising an N-terminal methionine-glycine dipeptide. In some embodiments, the nucleic acid molecule encoding the one or more Pol polyprotein components encodes the methionine-glycine dipeptide at the 5' end, and in some embodiments the nucleic acid molecule encoding the one or more Pol polyprotein components encodes the methionine-glycine dipeptide internally.

Some aspects of the invention provide that the one or more Pol polyprotein components (e.g., integrase) are fused

to a homing protein, which can direct the component, such as integrase, to a specific sequence in the host cell. In some embodiments the homing protein recognizes a palindromic sequence, and in some embodiments, the homing protein recognizes a restriction site. In some embodiments the homing protein is a restriction enzyme. In some embodiments the restriction enzyme is I-PpoI. In some embodiments the homing protein is a nuclease-inactivated restriction enzyme. In some embodiments, the homing protein is I-PpoI N119A.

Some aspects of the invention provide for increased translations in host cells. e.g., by providing that the one or more nucleic acid molecules of the composition are codon optimized for expression in the host cell. In some aspects, the host cell is a mammalian cell. In some aspects, the host cell is a human cell. In some aspects, the host cell is an avian cell.

Some aspects of the invention provide that the composition further comprises a priming oligonucleotide. In some embodiments, the priming oligonucleotide is GUCCCU-GUUCGGGCGCCA (SEQ ID NO: 18) or GTCCCTGTTTCGGGCGCCA (SEQ ID NO: 19). In some embodiments, the priming oligonucleotide may be an engineered sequence that is complementary to a reverse transcriptase priming element.

Some aspects of the invention provide that the nucleic acid molecules are RNA molecules or DNA molecules. For example, the nucleic acid molecules may be ssDNA molecules, dsDNA molecules, ssRNA molecules, or dsRNA molecules.

Some aspects of the invention provide that the nucleic acid molecule comprising the one or more transgenes can comprise two or more transgenes, where the transgenes are separated by one or more polycistronic elements. The polycistronic elements may be one or more IRESes and/or one or more sequences encoding a 2A peptide and/or other polycistronic elements. This nucleic acid molecule may comprise one or more enhancers. For example, the one or more enhancers may include a woodchuck hepatitis virus post-transcriptional regulatory element (WPRE).

Some aspects of the invention provide that the nucleic acid molecules are RNA molecules and comprise one or more modifications selected from modified ribonucleosides, a 5'-7mG cap structure, and a poly (rA) tail. In some embodiments, the modified ribonucleoside is pseudouridine or a derivative of pseudouridine (e.g., N1-methylpseudouridine). In some embodiments, the modified ribonucleoside is N6-methyladenosine. In other embodiments, the nucleic acid molecule comprising the one or more transgenes can be an RNA molecule and further comprise one or more modifications selected from a 5'-7mG cap structure and a poly (rA) tail.

Some aspects of the invention include that the Pol polyprotein components, accessory proteins, and/or LTRs are based on Pol polyprotein components, accessory proteins, and/or LTRs are from human immunodeficiency virus (HIV), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), human foamy virus (HFV), murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis

virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV), Friend MLV (FrMLV), Avian sarcoma virus (ASV), Avian leukosis virus, Avian myeloblastosis virus, UR2 sarcoma virus. Y73 sarcoma virus, Jaagsiekte sheep retrovirus, Langur virus, Mason-pfizer monkey virus. Squirrel monkey retrovirus, Avian carcinoma mill hill virus 2, Bovine leukemia virus, Primate T-lymphotropic virus 1, Primate T-lymphotropic virus 2. Primate T-lymphotropic virus 3, Walleye dermal sarcoma virus, Walleye epidermal hyperplasia virus 1. Walleye epidermal hyperplasia virus 2, Chick syncytial virus, Feline leukemia virus, Finkel-Biskis-Jenkins murine sarcoma virus, Gardner-Amstein feline sarcoma virus, Gibbon ape leukemia virus, Guinea pig type-C oncovirus. Hardy-Zuckerman feline sarcoma virus, Harvey murine sarcoma virus, Kirsten murine sarcoma virus, Koala retrovirus, Moloney murine sarcoma virus, Porcine type-C oncovirus, Reticuloendotheliosis virus, Snyder-Theilen feline sarcoma virus, Trager duck spleen necrosis virus, Viper retrovirus, Woolly monkey sarcoma virus, Jembrana disease virus, Puma lentivirus. Bovine foamy virus, Equine foamy virus, Feline foamy virus, Brown greater galago prosimian foamy virus, Bornean orangutan simian foamy virus, Central chimpanzee simian foamy virus, Cynomolgus macaque simian foamy virus, Eastern chimpanzee simian foamy virus, Grivet simian foamy virus. Guenon simian foamy virus, Japanese macaque simian foamy virus, Rhesus macaque simian foamy virus, Spider monkey simian foamy virus. Squirrel monkey simian foamy virus, Taiwanese macaque simian foamy virus, Western chimpanzee simian foamy virus, Western lowland gorilla simian foamy virus. White-tufted-ear marmoset simian foamy virus, Yellow-breasted capuchin simian foamy virus, or any combination thereof. Other aspects of the invention provide that the nucleic acid molecules are packaged in a non-viral delivery system. For example, they may be packaged in lipid nanoparticles.

Aspects of the invention provide that the one or more promoters comprise one or more tissue-specific or cell-specific promoters. For example, the promoters may be specific to bone marrow, hematopoietic stem cells (HSCs), epithelial cells, liver cells, ocular cells, muscle cells, or T cells. The one or more promoters may include an hCMV promoter.

Some aspects of the invention provide that the one or more transgenes encode one or more therapeutic proteins, diagnostic proteins, or reporter proteins, or fragments thereof. For example, the one or more transgenes may encode one or more therapeutic proteins, diagnostic proteins, or reporter proteins, or fragments thereof. The therapeutic protein could be beta-globin, cystic fibrosis transmembrane conductance regulator (CFTR), Factor VIII, dystrophin, or RP GTPase regulator (RPGR). The reporter protein could be a fluorescent protein or luciferase.

Other aspects of the invention provide that the non-viral delivery system is targeted to a specific tissue or cell type. The specific tissue or cell type could be bone marrow, HSCs, epithelial cells, liver cells, ocular cells, muscle cells, or T cells. The non-viral delivery system is a lipid nanoparticle, a liposome, a polypeptide nanoparticle, a silica nanoparticle, a gold nanoparticle, a polymeric nanoparticle, a dendrimer, a cationic nanoemulsion, an inorganic carrier (such as CaP), a polymer and lipid hybrid carrier.

Methods for expressing a gene in a subject in need thereof are also provided, where the method includes administering to the subject an effective amount of the above-referenced nucleic acids or a composition thereof (e.g., lipid nanoparticle) resulting in the expression of one or more transgenes

in the subject. The method results in expressing a gene in a cell by delivering the nucleic acids into the cell. The method could include using the nucleic acids or a composition thereof (e.g., lipid nanoparticle) by delivering them to a subject, thereby expressing the one or more transgenes in the subject.

Aspects of the invention provide methods of treating a disease or condition in a subject in need thereof by delivering the nucleic acids or a composition thereof (e.g., lipid nanoparticle) to the subject, thereby expressing the one or more transgenes in the subject. The disease or condition to be treated can be a genetic disease or condition. The disease or condition can be a hereditary genetic disease or condition. The disease or condition can be sickle cell disease, beta-thalassemia, haemophilia B, retinitis pigmentosa, Duchenne muscular dystrophy, cystic fibrosis, or cancer.

Some aspects of the invention provide that the one or more transgenes are integrated into the genome of a target cell. For example, the one or more transgenes may be stably expressed for at least a week, at least two weeks, at least a month, at least 6 months, at least a year, or for the lifetime of the subject.

Some aspects of the invention provide methods of eliciting an immune response in a subject in need thereof by administering to the subject an effective amount of the nucleic acids or a composition thereof (e.g., lipid nanoparticle), thereby expressing the one or more transgenes in the subject. For example, the subject can have cancer, and the one or more transgenes encode a tumor antigen. Alternatively, the subject can have or be at risk of contracting an infectious disease, and the one or more transgenes encode an antigen associated with the infectious disease. Embodiments include delivery of the nucleic acids locally or systemically.

Aspects of the invention also provide one or more nucleic acid templates comprising a 5' UTR, a nucleic acid sequence encoding one or more retroviral Pol polyprotein components, and a 3' UTR, wherein the expression of Pol polyprotein components do not require translational slippage from an in-line gag gene. Other aspects of the invention provide one or more nucleic acid templates comprising a 5' UTR, a nucleic acid sequence encoding a gag-pol gene comprising a frameshift mutation, and a 3' UTR. Yet other aspects of the invention provide one or more nucleic acid templates comprising a 5' UTR, a nucleic acid sequence encoding a gag-pol gene, and a 3' UTR, wherein the gag-pol gene does not encode the matrix protein. The nucleic acid templates can also include a nucleic acid sequence encoding one or more accessory proteins selected from MA, p6, NC, CA, Vif, Tat, Nef, Vpr, and Vpu. In some embodiments, the Pol polyprotein components and/or accessory proteins of the templates are based on Pol polyproteins and accessory proteins from either human immunodeficiency virus (HIV), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), human foamy virus (HFV), murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV), Friend MLV (FrMLV), Avian sarcoma virus (ASV), Avian leukosis virus, Avian myeloblastosis virus, UR2 sarcoma virus, Y73 sar-

coma virus, Jaagsiekte sheep retrovirus. Langur virus. Mason-pfizer monkey virus, Squirrel monkey retrovirus, Avian carcinoma mill hill virus 2, Bovine leukemia virus, Primate T-lymphotropic virus 1, Primate T-lymphotropic virus 2, Primate T-lymphotropic virus 3, Walleye dermal sarcoma virus, Walleye epidermal hyperplasia virus 1, Walleye epidermal hyperplasia virus 2, Chick syncytial virus, Feline leukemia virus, Finkel-Biskis-Jenkins murine sarcoma virus, Gardner-Amstein feline sarcoma virus, Gibbon ape leukemia virus, Guinea pig type-C oncovirus, Hardy-Zuckerman feline sarcoma virus, Harvey murine sarcoma virus, Kirsten murine sarcoma virus, Koala retrovirus, Moloney murine sarcoma virus, Porcine type-C oncovirus, Reticuloendotheliosis virus, Snyder-Theilen feline sarcoma virus, Trager duck spleen necrosis virus, Viper retrovirus, Woolly monkey sarcoma virus, Jembrana disease virus. Puma lentivirus, Bovine foamy virus, Equine foamy virus, Feline foamy virus, Brown greater galago prosimian foamy virus, Bornean orangutan simian foamy virus, Central chimpanzee simian foamy virus. Cynomolgus macaque simian foamy virus, Eastern chimpanzee simian foamy virus, Grivet simian foamy virus, Guenon simian foamy virus. Japanese macaque simian foamy virus, Rhesus macaque simian foamy virus, Spider monkey simian foamy virus, Squirrel monkey simian foamy virus, Taiwanese macaque simian foamy virus, Western chimpanzee simian foamy virus, Western lowland gorilla simian foamy virus, White-tufted-ear marmoset simian foamy virus, Yellow-breasted capuchin simian foamy virus, or any combination thereof. Methods of producing a nucleic acid, such as an RNA, are also provided, where the method includes the in vitro transcription of the nucleic acid template.

Other aspects of the invention provide a kit comprising the nucleic acids or nucleic acid template mentioned above. The nucleic acids or nucleic acid templates may be provided in one or more containers.

Other advantages, features, and uses of the invention will be apparent from the detailed description of certain exemplary, non-limiting embodiments, the drawings; the non-limiting working examples; and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic which outlines the approach for an mRNA-based gene therapy platform.

FIG. 2 is a table which describes examples of current approaches to gene therapy and the major advantages and challenges associated with each approach.

FIG. 3 is a schematic which outlines an exemplary embodiment of the transgene-encoding transfer nucleic acid design. RNA production elements include an RNA polymerase promoter (e.g., the T7 RNA polymerase promoter) and a restriction digestion site, for template linearization.

FIG. 4 is a schematic which illustrates an exemplary embodiment of a reverse transcription and integration complex design, compared to the canonical design. The Pol polyprotein is produced from a separate construct as compared to the inline gag/pol design from the canonical design so translational slippage is no longer required. In some embodiments, one or more accessory retroviral proteins are encoded in cis with the Pol polyprotein. RNA production elements include an RNA polymerase promoter (e.g., the T7 RNA polymerase promoter) and optionally, one or more restriction digestion sites for template linearization.

FIGS. 5A-5C illustrate an exemplary embodiment of a reverse transcription and integration machinery design, wherein the Pol polyprotein components are encoded as

separate proteins on one or more nucleic acid molecule. RNA production elements include an RNA polymerase promoter (e.g., the T7 RNA polymerase promoter) and optionally, one or more restriction digestion sites for template linearization. FIG. 5A is a schematic which illustrates an exemplary embodiment in which the protease (PR), reverse transcriptase/RNase H (RT) and integrase (IN) are each encoded on a single nucleic acid construct and where each component is separated by autoprocessing by the PR. FIG. 5B is a schematic which illustrates an exemplary embodiment in which the individual units RT and IN are on a bicistronic construct. FIG. 5C is a schematic which illustrates an exemplary embodiment in which the individual units RT and IN are on separate constructs.

FIG. 6 is another schematic depicting an mRNA-based gene therapy approach. The viral machinery required for reverse transcription and integration (e.g., viral enzymes and/or one or more accessory proteins) and the transgene are provided on separate RNAs. A non-viral delivery system, optionally using surface markers, is used to deliver the RNAs to the target cell. The transgene-encoding RNA is also referred to as the lentivector 'genome' RNA. The expression of the transgene relies on the translation of the viral enzymes, conversion of the lentivector 'genome' RNA into DNA, and integration of the DNA into genomic DNA in the target cell, which results in long-lasting expression of the transgene.

FIGS. 7A and 7B show that RNA-derived functional units drive reporter gene expression from transgene-encoding RNAs. FIG. 7A shows a schematic of the functional units, a Gag polyprotein RNA and a Pol polyprotein RNA (the functional unit mRNAs), and a transgene-encoding RNA (the 'genome' RNA). FIG. 7B shows the expression of the reporter transgene when transfected with the functional unit RNAs.

FIGS. 8A-8D show that optimized functional unit RNAs demonstrate enhanced expression and localize to target organelles. FIG. 8A is a schematic depicting RNA constructs encoding WT integrase (IN); stabilized Met-Gly, human codon optimized (hCO) IN; and stabilized Met-Gly hCO ΔINS IN. FIG. 8B shows a capillary electropherogram of Met-Gly, hCO ΔINS IN mRNA before and after polyadenylation. FIG. 8C shows a Western blot analysis of the expression of WT IN mRNA, Met-Gly hCO IN mRNA; and Met-Gly hCO ΔINS IN mRNA. FIG. 8D shows confocal microscopy images showing the subcellular localization of optimized integrase following RNA transfection into 293FT cells.

FIG. 9 illustrates an exemplary embodiment where the reverse transcriptase is encoded as individual p51 and p66 subunits.

FIG. 10 illustrates an exemplary embodiment where integrase is fused to a homing protein for site-directed genome integration. Integrase may be fused to a homing protein when expressed as a part of the Pol polyprotein or when expressed as an individual unit.

DETAILED DESCRIPTION

According to some aspects of the present disclosure, minimal design nucleic acid compositions are effective for in vivo gene therapy. This minimal design can overcome some of the limitations of the lentiviral approach, including the need to produce virus by expansion in cells, which is slow and expensive; and the need for processing of viral particles, which can be inefficient. These are examples of the advantages of the disclosed nucleic acid compositions, and

those skilled in the art will recognize the other advantages. The disclosure also provides nucleic acid molecules, methods of use, methods of treatment, nucleic acid templates, cells, and kits. The disclosure also provides for modifications to increase the stability and/or functioning of the nucleic acid, and/or the stability and/or functioning of the encoded protein. These modifications can include mutations in one or more intrinsic instability (INS) elements; a functional unit, such as an integrase, with an N-terminal methionine-glycine dipeptide; fusion of a functional unit, such as an integrase, to a homing protein; use of codon optimization for expression in a host cell; and/or use of a priming oligonucleotide. Other modifications are as disclosed herein.

Provided herein, in some aspects, are nucleic acids and compositions of nucleic acids for the delivery of a transgene (for example, but not limited to beta-globin, cystic fibrosis transmembrane conductance regulator (CFTR), Factor VIII, dystrophin, or RP GTPase regulator (RPGR)), kits, and methods of use. In some aspects, the disclosure provides non-viral compositions based on retroviruses (e.g., lentiviruses) comprising: at least a first nucleic acid molecule encoding a retroviral Pol polyprotein which is processed to provide for protease activity, reverse transcription, and integration of a transgene-encoding nucleic acid molecule, and at least a second nucleic acid molecule comprising one or more transgenes flanked by long terminal repeat (LTR) sequences, for delivery of one or more transgenes to a target cell. The nucleic acid molecule comprising one or more transgenes may also be referred to herein as "the transgene-encoding nucleic acid molecule." In some aspects, the disclosure provides for the encoding of the components of the Pol polyprotein as separate, multiple functional units, including, but not limited to, the units encoding reverse transcription and integration activity. These multiple functional units can be encoded on one nucleic acid or two or more nucleic acids. The protease of the Pol polyprotein can also be encoded on the same or a different construct as one or more of the other Pol polyprotein components or omitted if the required enzymatic activities (e.g., reverse transcriptase, integrase) are not encoded as viral polyproteins. In some embodiments, the Pol polyprotein components are encoded on one or more nucleic acids with intervening polycistronic elements. The polycistronic elements may be IRESes and/or 2A sequences and/or other polycistronic elements. In some embodiments, the compositions of the disclosure further comprise one or more nucleic acid sequences encoding one or more accessory retroviral proteins (accessory proteins) which may enhance reverse transcription and integration of the transgene-encoding nucleic acid. As used herein, an "accessory protein" refers to any retroviral protein other than a Pol polyprotein component that can enhance reverse transcription and/or integration of the transgene-encoding nucleic acid. An accessory protein may be a structural protein, including, but not limited to, MA, CA, NC, and Env, or a non-structural protein, including, but not limited to, Tat, Rev, Nef, Vpr, Vpu, and Vif. The nucleic acid sequences encoding the one or more accessory retroviral proteins can be on one or more nucleic acids encoding the Pol polyprotein components or can be on one or more nucleic acids that do not encode the Pol polyprotein components. In some embodiments, the stability of the mRNAs is increased by, for example, introducing mutations in one or more intrinsic instability (INS) elements. In some embodiments, the stability of the translated proteins is increased by, for example, incorporating an N-terminal methionine-glycine dipeptide. In some embodiments, the translation efficiency of the mRNA is increased using codon

optimization for expression in the host cell (e.g., codon optimization for expression in human cells). In some embodiments, the proteins of the disclosure, such as integrase, are fused to a homing protein. In some embodiments, the composition comprises a priming oligonucleotide. The nucleic acids and compositions of the disclosure may be used for ex vivo or in vivo expression of one or more transgenes by a cell, tissue, organ, or subject, or in vitro for the production of recombinant proteins. In addition, the nucleic acid molecules can be packaged into a non-viral delivery system (e.g., a lipid nanoparticle), thus avoiding the need to produce viral particles, which requires expansion in cells and is slow and expensive, and avoiding the risks associated with virus-based approaches such as immunogenicity.

Definitions

General methods in molecular and cellular biochemistry can be found in such textbooks as *Molecular Cloning: A Laboratory Manual*, 3rd Ed. (Sambrook et al., CSH Laboratory Press 2001); *Short Protocols in Molecular Biology*, 4th Ed. (Ausubel et al. eds., John Wiley & Sons 1999); *Protein Methods* (Bollag et al., John Wiley & Sons 1996); *Nonviral Vectors for Gene Therapy* (Wagner et al. eds., Academic Press 1999); *Viral Vectors* (Kaplift & Loewy eds., Academic Press 1995); *Immunology Methods Manual* (I. Lefkovits ed., Academic Press 1997); and *Cell and Tissue Culture: Laboratory Procedures in Biotechnology* (Doyle & Griffiths, John Wiley & Sons 1998), the disclosures of which are incorporated herein by reference.

An antibody (interchangeably used in plural form) is an immunoglobulin molecule capable of specific binding to a target, such as a carbohydrate, polynucleotide, lipid, polypeptide, etc., through at least one antigen recognition site, located in the variable region of the immunoglobulin molecule. As used herein, the term “antibody” encompasses not only intact (e.g., full-length) polyclonal or monoclonal antibodies, but also antigen-binding fragments thereof (such as Fab, Fab', F(ab')₂, Fv), single chain (scFv), mutants thereof, fusion proteins comprising an antibody portion, humanized antibodies, chimeric antibodies, diabodies, nanobodies, linear antibodies, single chain antibodies, and any other modified configuration of the immunoglobulin molecule that comprises an antigen recognition site of the required specificity, including glycosylation variants of antibodies, amino acid sequence variants of antibodies, and covalently modified antibodies.

An “expression cassette” or “nucleic acid template” comprises any nucleic acid construct capable of serving as a template for the production of DNA or RNA, including a gene/coding sequence of interest as well as non-translated RNAs.

A “polyprotein” refers to a polypeptide comprising covalently conjoined smaller proteins that can occur in nature and is typically cleaved into the constituent proteins with different biological functions.

The “Gag polyprotein” refers to the precursor polyprotein encoded by the gag (group-specific antigen) gene. The Gag polyprotein is processed during maturation to form matrix protein (MA), capsid protein (CA), spacer peptide 1 (SP1), nucleocapsid protein p7 (NC), spacer peptide 2 (SP2), and P6 protein. The terms “Gag protein” and “Gag polyprotein component” refers to one or more of the proteins encoded by the gag gene, i.e., MA, CA, SP1, NC, SP2, or P6.

The “Pol polyprotein” refers to the precursor polyprotein encoded by the pol gene. The Pol polyprotein is processed

to form protease (PR), reverse transcriptase/RNase H (RT/RH), and integrase (IN). Each component is separated by autoprocessing by PR. The terms “Pol protein” and “Pol polyprotein component” refer to one or more of the proteins encoded by the pol gene, i.e., RT/RH, IN, or PR.

The “percent identity” of two nucleic acid sequences may be determined by any method known in the art. In some embodiments, the percent identity of two nucleic acid sequences is determined using the algorithm of Karlin and Altschul, *Proc. Natl. Acad. Sci. USA* 87:2264-68, 1990, modified as in Karlin and Altschul, *Proc. Natl. Acad. Sci. USA* 90:5873-77, 1993. Such an algorithm is incorporated into the NBLAST and XBLAST programs (version 2.0) of Altschul et al., *J. Mol. Biol.* 215:403-10, 1990. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12, to obtain guide sequences homologous to a target nucleic acid. Where gaps exist between two sequences, Gapped BLAST can be utilized as described in Altschul et al., *Nucleic Acids Res.* 25(17):3389-3402, 1997. When utilizing BLAST and Gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used.

The terms “polycistronic” and “multicistronic” are used interchangeably to refer to a nucleic acid molecule that encodes more than one polypeptide separately within the same nucleic acid molecule. For example, a polycistronic mRNA is an mRNA that encodes more than one polypeptide in the same transcript. A “polycistronic element” or “multicistronic element” refers to an element that separates individual polypeptides in a polycistronic nucleic acid to allow for independent translation and/or post-translational processing.

The terms “polypeptide” and “protein” are used interchangeably to refer to a polymer of amino acid residues linked by peptide bonds, and for the purposes of the instant disclosure, have a minimum length of at least 5 amino acids. Both full-length proteins and fragments thereof greater than 5 amino acids are encompassed by the definition. The terms also include polypeptides that have co-translational (e.g., signal peptide cleavage) and post-translational modifications of the polypeptide, such as, for example, disulfide-bond formation, glycosylation, acetylation, phosphorylation, proteolytic cleavage (e.g., cleavage by furins or metalloproteases), and the like. Furthermore, as used herein, a “polypeptide” or “protein” refers to a protein that includes modifications, such as deletions, additions, and substitutions (generally conservative in nature as would be known to a person in the art) to the native sequence, as long as the protein maintains the desired activity relevant to the purposes of the described methods. These modifications can be deliberate, as through site-directed mutagenesis, or can be accidental, such as through mutations of hosts that produce the proteins, or errors due to PCR amplification or other recombinant DNA methods.

The terms “nucleic acid” or “nucleic acid molecule,” as used herein, generally refers to deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). In some embodiments, “nucleic acid” refers to individual nucleic acid residues (e.g., nucleotides and/or nucleosides). In some embodiments, “nucleic acid” refers to an oligonucleotide chain comprising individual nucleotides. As used herein, the terms “oligonucleotide” and “polynucleotide” can be used interchangeably to refer to a polymer of nucleotides (e.g., a string of at least two nucleotides). A nucleic acid may be single-stranded or double-stranded. The nucleotide monomers in the nucleic acid molecules may be naturally occurring

nucleotides, modified nucleotides or combinations thereof. Modified nucleotides, in some embodiments, comprise modifications of the sugar moiety and/or the pyrimidine or purine base.

The terms “subject,” and “patient,” are used interchangeably herein. In some embodiments, a subject is a mammal. “Mammal” refers to any animal classified as a mammal, including humans, domestic and farm animals, and zoo, sports, or pet animals, such as dogs, horses, cats, cows, sheep, goats, pigs, camels, etc. In some embodiments, the mammal is human. In some embodiments, the subject is avian.

A “therapeutically effective amount” of the compositions of the disclosure generally refers to an amount sufficient to elicit the desired biological response, e.g., express the transgene in a target cell, treat the condition, etc. As will be appreciated by those of ordinary skill in this art, the effective amount of an agent described herein may vary depending on such factors as the condition being treated, the mode of administration, and the age, body composition, and health of the subject.

The term “transgene” or “transgene sequence” refers to an exogenous nucleotide sequence encoding a protein or RNA product of interest.

The terms “treat”, “treating”, “treatment”, and “therapy” encompass an action that occurs while a subject is suffering from a condition which reduces the severity of the condition (or a symptom associated with the condition) or retards or slows the progression of the condition (or a symptom associated with the condition).

Minimal Gene Therapy Platform

The disclosure is based in part on the finding that nucleic acid molecules of retroviral (e.g., lentiviral) origin can be used to deliver, integrate, and stably express transgenes (e.g., for use in gene therapy) using a minimal system comprising at least a first nucleic acid encoding a retroviral Pol polyprotein which is processed to provide for, potentially among other things, reverse transcription and integration of a transgene-encoding nucleic acid, and at least a second nucleic acid molecule comprising one or more transgenes flanked by LTR sequences. In some embodiments, the components of the Pol polyprotein are encoded as separate units on one or more nucleic acid molecules. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the system comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some embodiments, a minimal system comprises two nucleic acid molecules, one nucleic acid molecule providing the reverse transcriptase and integrase functions (either as a polyprotein that is processed with the protease, or as separately encoded units), and one nucleic acid molecule encoding the one or more transgenes. In some embodiments, the minimal system comprises two or more nucleic acid molecules (e.g., 2, 3, 4, or 5, or more). The nucleic acid molecules can be packaged into a non-viral delivery system (e.g., a lipid nanoparticle), thus bypassing the production of viral particles, which requires expansion in cells and is slow and expensive, and avoiding the risks associated with virus-based approaches, such as immunogenicity.

The compositions of the disclosure have several advantages over current gene and cell therapy approaches. Some of the challenges associated with current gene therapy approaches, such as CRISPR/Cas9, retroviruses, lentiviruses, herpesviruses, adenoviruses, and adeno-associated

viruses (AAV), are shown in FIG. 2. Compared to existing approaches, the compositions of the disclosure can function in both dividing and non-dividing cells, have high transgene size capacity (e.g., 4-5 kb, 5-6 kb, 6-7 kb, 7-8 kb, 8-9 kb, 9-10 kb, or greater than 4 kb, greater than 5 kb, greater than 6 kb, greater than 7 kb, greater than 8 kb, greater than 9 kb, or greater than 10 kb), and/or do not require the costly production of viruses and/or viral particles. The compositions of the disclosure also overcome some of the limitations associated with HIV-1 and third generation lentiviral vectors. In HIV-1 vectors, all of the components required for the HIV-1 life cycle are on a single RNA genome. HIV-1 vectors are also replication competent (i.e., it can replicate after integration), have a packaging signal, which allows for packaging the genome into a viral particle, have an mRNA nuclear export element for HIV-1 mRNA, and have viral machinery in cis. HIV-1 vectors present safety concerns that preclude their in vivo use. Next generation lab-based lentiviruses, on the other hand, feature self-inactivation after the integration step, viral machinery in trans on separate constructs, and transgenes with promoter/enhancer regions. However, these require co-transfection of packaging cells with multiple plasmids, resulting in inefficient viral particle production. Partially packaged or mispackaged viral particles are also highly immunogenic. In an improvement upon the aforementioned lentiviral vectors, the minimal RNA design described herein does not require particle formation or packaging, and in some embodiments, features only reverse transcription and integration functions in trans on a separate construct. By requiring only minimal components providing reverse transcription and integration and transfer (i.e., transgene-encoding) functions, while being self-inactivating and bypassing the production of viral particles, the compositions of the disclosure address the challenges associated with current lentiviral approaches. Further, non-viral delivery systems can be easily functionalized with targeting moieties, simplifying administration and nucleic acid synthesis (e.g., RNA synthesis) and delivery system production can both be cell-free, thus simplifying the manufacturing process.

The compositions described herein comprise the following basic components: one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription, for example RT/RH, and integration, for example IN, of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising one or more transgene sequences flanked by LTR sequences. The Pol polyprotein can be expressed and processed to provide protease activity (for example with the PR protein), reverse transcriptase activity (for example with the RT/RH protein), and integration (for example with the IN protein) of the transgene-encoding nucleic acid molecule. The Pol polyprotein components can instead be expressed as separate, multiple functional units on one or more nucleic acid molecules. When expressed as separate units, the Pol polyprotein component providing protease activity may or may not be expressed. The compositions of the disclosure may optionally further comprise one or more nucleic acid sequences encoding one or more accessory retroviral proteins to enhance reverse transcription and integration of the transgene-encoding nucleic acid. The one or more accessory proteins may be encoded in cis or in trans with the one or more Pol polyprotein components. In some embodiments, the nucleic acid molecules encoding the Pol polyprotein further encode one or more accessory retroviral proteins. In some embodiments, the compositions of the disclosure comprise one or more additional nucleic acid molecules

encoding one or more accessory retroviral proteins. In some embodiments, a composition comprises two nucleic acid molecules, one nucleic acid molecule providing the reverse transcriptase and integrase functions (for example, but not limited to Pol polyprotein and/or one or more accessory retroviral proteins) and one nucleic acid molecule encoding the one or more transgenes. In some embodiments, a composition comprises two or more nucleic acid molecules (e.g., 2, 3, 4, or 5, or more). In some embodiments, a composition does not comprise any accessory retroviral proteins. The compositions of the disclosure are packaged in a non-viral delivery system (e.g., a lipid nanoparticle) for in vivo or ex vivo delivery.

The nucleic acid molecules may be RNA molecules or DNA molecules, both genomic and/or cDNA, or a hybrid of RNA and DNA, wherein the nucleic acid contains any combination of deoxyribonucleotides and ribonucleotides (e.g., artificial or natural). The nucleic acid molecules may be single-stranded (ss) or double-stranded (ds), or may contain portions of both single-stranded and double-stranded sequence. In some embodiments, the nucleic acid molecules are DNA molecules. The DNA molecules may be ssDNA molecules or dsDNA molecules (e.g., linear dsDNA molecules). In some embodiments, the nucleic acid molecules are RNA molecules. The RNA molecules may be ssRNA or dsRNA molecules. In some embodiments, the RNA molecules may be ssRNA molecules. In some embodiments, the nucleic acid molecules may be mRNA molecules.

In some embodiments, a composition of the disclosure comprises one or more RNA molecules encoding retroviral Pol polyprotein components, and an RNA molecule comprising one or more transgene sequences flanked by LTR sequences. In some embodiments, a composition of the disclosure comprises two RNA molecules: a first RNA molecule encoding a retroviral Pol polyprotein and a second RNA molecule comprising a transgene sequence flanked by LTR sequences. In some embodiments, the RNA molecule encoding the Pol polyprotein expresses the polyprotein as a canonical unit with each component separated by protease processing. In some embodiments, the RNA molecule encoding the Pol polyprotein components expresses them as individual units, for example, as a bicistronic construct. In some embodiments, the RNA molecule encoding the Pol polyprotein further encodes one or more accessory retroviral proteins as described herein. In some embodiments, the composition comprises one or more additional RNA molecules encoding one or more accessory retroviral proteins. The in vivo approach for embodiments in which the nucleic acid molecules are RNA molecules is outlined in FIG. 1 and FIG. 6.

Reverse Transcription and Integration Functions

One or more nucleic acids of the system encode the retroviral Pol polyprotein which is processed to provide for the functional units responsible for protease activity, reverse transcription, and integration of the one or more transgenes. In some embodiments, a nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding a retroviral Pol polyprotein (the retroviral pol gene), and a 3' UTR, provides the reverse transcription and integration functions. Thus, in some embodiments, all Pol polyprotein components are encoded on one nucleic acid molecule with a 5' and 3' UTR. In some embodiments, the Pol polyprotein components are encoded on two or more nucleic acid molecules each with a 5' and 3' UTR. In some embodiments, the Pol polyprotein components are encoded on one or more nucleic acid molecules each with a 5' and 3' UTR and are separated by one or more polycistronic elements (e.g., one or more

IREs and/or 2A peptide-encoding sequences, and/or other polycistronic elements). Any other polycistronic element that will facilitate translation could also be used. In some embodiments, the nucleic acid molecules do not encode the Pol protease, e.g., where the Pol polyprotein components reverse transcriptase and integrase are encoded as separate units on one or more nucleic acid molecules or are separated by intervening polycistronic elements (e.g., IREs and/or 2A peptide-encoding sequences, and/or other polycistronic elements). In further embodiments, the Pol polyprotein component reverse transcriptase may be encoded as individual p51 and p66 subunits (FIG. 9). In some embodiments, p51 and p66 are encoded on separate nucleic acid molecules each with a 5' and 3' UTR. In some embodiments, p51 and p66 are encoded as separate units on the same nucleic acid separated by a polycistronic element (e.g., an IRES or 2A peptide-encoding sequence, or other polycistronic element).

Compared to the canonical design where translational slippage from gag is required for pol expression, in some embodiments, expression of Pol proteins can be increased by producing the Pol polyprotein or the individual Pol polyprotein components from a separate construct and by encoding different UTRs for mRNA stability and high levels of translation. In some embodiments, heterologous UTRs are selected to improve mRNA stability and/or level of translation. In some embodiments, the UTRs include 3' or 5' sequences from mRNA molecules which are stable and highly translated (for example, but not limited to, β -globin, actin, GAPDH, tubulin, histone, or citric acid cycle enzymes). Engineered or synthetic UTRs (i.e., UTRs based on rational design that are non-naturally occurring and distinct from mutated naturally-occurring UTRs) may also be used to enhance stability and/or translation. The sequences of human UTRs are available on GenBank.

In some embodiments, in addition to encoding functional units, the nucleic acid sequences encode one or more accessory retroviral proteins to enhance reverse transcription and/or integration. In some embodiments, the one or more accessory proteins are encoded in cis (i.e., encoded by the same nucleic acid molecule) with one or more Pol polyprotein components. In some embodiments, the one or more accessory proteins are encoded in cis with all Pol polyprotein components. In some embodiments, the one or more accessory proteins are encoded in cis with the Pol polyprotein. Thus, in some embodiments, all Pol polyprotein components and accessory proteins are encoded by one nucleic acid molecule with a 5' and 3' UTR. In some embodiments, the nucleotide sequences encoding the Pol polyprotein components and/or the accessory proteins are separated by one or more polycistronic elements (e.g., one or more IREs and/or 2A peptide-encoding sequences, and/or other polycistronic elements). In some embodiments, the nucleotide sequences encoding the Pol polyprotein components and/or the accessory proteins encode a fusion protein comprising the Pol polyprotein components and/or the accessory proteins. In some embodiments, the one or more accessory proteins are encoded in trans (i.e., encoded by a separate nucleic acid molecule) from the Pol polyprotein components. In some embodiments, the Pol polyprotein components and accessory proteins are encoded by two or more (e.g., two, three, four, or five, or more) nucleic acid molecules, each with a 5' and 3' UTR. In some embodiments, the UTRs include 3' or 5' sequences from mRNA molecules which are stable and highly translated (including, but not limited to, globin, actin, GAPDH, tubulin, histone, or citric acid cycle enzymes).

In some embodiments, the one or more accessory proteins are selected from gag matrix protein (MA) (p17), nucleocapsid (NC) (p9), capsid protein (CA) (p24), p6, viral infectivity factor (Vif), transactivator-of-transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu). In some embodiments, the accessory proteins may be wild-type proteins or may be mutated proteins. In some embodiments, a mutated accessory protein is CAN74D or CA E45A. The numbering corresponds to the wild-type HIV-1 CA protein and corresponding mutations in other retroviral CA proteins can be readily determined by one of skill in the art. In some aspects, the wild-type HIV-1 strain can be NL4-3. In some embodiments, the system comprises one accessory protein selected from MA, NC, CA, p6, Vif, Tat, Nef, Vpr, and Vpu. In some embodiments, the accessory protein is NC. In some embodiments, the accessory protein is CA. In some embodiments, a composition of the disclosure comprises two accessory proteins selected from MA, NC, CA, p6, Vif, Tat, Nef, Vpr, and Vpu. In some embodiments, a composition of the disclosure comprises three accessory proteins selected from MA, NC, CA, p6, Vif, Tat, Nef, Vpr, and Vpu. In some embodiments, a composition of the disclosure comprises four, five, six, seven, or eight accessory proteins selected from MA, NC, CA, p6, Vif, Tat, Nef, Vpr, and Vpu. In some embodiments, the system comprises MA, NC, CA, p6, Vif, Tat, Nef, Vpr, and Vpu.

In some embodiments, the one or more accessory proteins are expressed from the gag polyprotein (p55). Thus, in some embodiments, a nucleic acid molecule comprises the retroviral gag gene. In some embodiments, a nucleic acid molecule of the disclosure comprises a nucleic acid sequence comprising the gag and pol genes in the canonical gag-pol overlapping orientation. In some embodiments, a nucleic acid molecule of the disclosure comprises a nucleic acid sequence encoding a fusion of the Gag and Pol polyproteins. In some embodiments, a nucleic acid of the disclosure comprises a nucleic acid sequence encoding Gag and Pol polyproteins separated by a 2A peptide-encoding sequence. In some embodiments, a nucleic acid molecule of the disclosure encodes one or more Pol polyprotein components and one or more Gag polyprotein components as separate units. In some embodiments, the separate units are joined by one or more polycistronic elements (e.g., one or more 2A peptide-encoding sequences and/or IRESes and/or other polycistronic elements). In some embodiments, a nucleic acid of the disclosure comprises a modified gag-pol gene that does not encode the matrix protein (MA). In some embodiments, a nucleic acid of the disclosure comprises a modified gag-pol gene comprising a frameshift mutation. In some embodiments, a nucleic acid of the disclosure comprises a modified gag-pol gene comprising a single nucleotide insertion disrupting the gag-pol frameshift mechanism. In some embodiments, a nucleic acid of the disclosure comprises a gag gene that does not encode MA.

In some embodiments, the nucleic acid molecules or compositions of the disclosure are capable of integrating the one or more transgenes into the host genome in the absence of functional retroviral Rev and/or Env proteins. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not contain nucleic acid sequences that express proteins encoded by at least one of: retroviral rev and retroviral env. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not contain nucleic acid sequences that express proteins encoded by both retroviral rev and env genes. In some embodiments, the nucleic acid molecules or compositions of the disclosure do

not comprise any nucleic acid sequences encoding retroviral Gag proteins. In other embodiments, the nucleic acid molecules or compositions of the disclosure do not comprise nucleic acid sequences that are able to express all retroviral Gag proteins. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not comprise nucleic acid sequences encoding the following retroviral Gag proteins: MA, SP1, SP2, and p6. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not comprise nucleic acid sequences encoding MA. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not comprise any nucleic acid sequences encoding retroviral Gag proteins. In other embodiments, the nucleic acid molecules or compositions of the disclosure do not comprise nucleic acid sequences that are able to express all retroviral Gag proteins.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding NC. In some embodiments, NC is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding NC in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes NC in cis. In some embodiments, nucleic acid sequences encoding other Gag proteins (including MA, CA, SP1, SP2, and p6) are not present. In some embodiments, nucleic acid sequences encoding MA, SP1, SP2, and p6 are not present.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding CA. In some embodiments, CA is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding CA in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes CA in cis. In some embodiments, nucleic acid sequences encoding other Gag proteins (including MA, NC, SP1, SP2, and p6) are not present. In some embodiments, nucleic acid sequences encoding MA, SP1, SP2, and p6 are not present.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding Vif. In some embodiments, Vif is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding Vif in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes Vif in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding Tat. In some embodiments, Tat is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding Tat in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes Tat in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding Nef. In some embodiments, Nef is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding Nef in cis. In

some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes Nef in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding Vpr. In some embodiments, Vpr is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding Vpr in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes Vpr in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding Vpu. In some embodiments, Vpu is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding Vpu in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes Vpu in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding MA. In some embodiments, MA is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding MA in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes MA in cis.

In some embodiments, a composition of the disclosure further comprises a nucleic acid sequence encoding p6. In some embodiments, p6 is encoded by a separate nucleic acid molecule comprising a 5' UTR and a 3' UTR. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further comprise a nucleic acid sequence encoding p6 in cis. In some embodiments, a nucleic acid molecule encoding a Pol polyprotein encodes p6 in cis.

In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from one or more retroviruses. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from the same retrovirus. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from two or more different retroviruses. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from the murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV), avian sarcoma leukosis virus (ASLV), endogenous xenotropic MuLV-related virus VP62 (XMRV), primate T-lymphotropic virus (PTLV), walleye dermal sarcoma virus (WDSV), the human foamy virus (HFV), Friend MLV (FrMLV), Avian sarcoma virus (ASV), Avian leukosis virus, Avian myeloblastosis virus, UR2 sarcoma virus, Y73 sarcoma virus, Jaagsiekte sheep retrovirus, Langur virus, Mason-pfizer monkey virus, Squir-

rel monkey retrovirus, Avian carcinoma mill hill virus 2, Bovine leukemia virus, Primate T-lymphotropic virus 1, Primate T-lymphotropic virus 2, Primate T-lymphotropic virus 3, Walleye dermal sarcoma virus, Walleye epidermal hyperplasia virus 1, Walleye epidermal hyperplasia virus 2, Chick syncytial virus, Feline leukemia virus, Finkel-Biskis-Jenkins murine sarcoma virus, Gardner-Amstein feline sarcoma virus, Gibbon ape leukemia virus, Guinea pig type-C oncovirus, Hardy-Zuckerman feline sarcoma virus, Harvey murine sarcoma virus, Kirsten murine sarcoma virus, Koala retrovirus, Moloney murine sarcoma virus, Porcine type-C oncovirus, Reticuloendotheliosis virus, Snyder-Theilen feline sarcoma virus, Trager duck spleen necrosis virus, Viper retrovirus, Woolly monkey sarcoma virus, Jembrana disease virus, Puma lentivirus, Bovine foamy virus, Equine foamy virus, Feline foamy virus, Brown greater galago prosimian foamy virus, Bornean orangutan simian foamy virus, Central chimpanzee simian foamy virus, Cynomolgus macaque simian foamy virus, Eastern chimpanzee simian foamy virus, Grivet simian foamy virus, Guenon simian foamy virus, Japanese macaque simian foamy virus, Rhesus macaque simian foamy virus, Spider monkey simian foamy virus, Squirrel monkey simian foamy virus, Taiwanese macaque simian foamy virus, Western chimpanzee simian foamy virus, Western lowland gorilla simian foamy virus, White-tufted-ear marmoset simian foamy virus, Yellow-breasted capuchin simian foamy virus, or a combination thereof. In some embodiments, the nucleic acid sequences are based on nucleic acid sequences from HFV. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from one or more lentiviruses. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from: human immunodeficiency virus (HIV) (e.g., HIV-1, HIV-2), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), jembrana disease virus, puma lentivirus, or a combination thereof. In some embodiments, the nucleic acid sequences are based on nucleic acid sequences from HIV-1 (e.g., GenBank Accession No. AF033819). Nucleotide sequences from the HIV genome are available to the skilled artisan. See, e.g., HIV Genome Browser (Los Alamos National Laboratory): www.hiv.lanl.gov/content/sequence/genome_browser/browser and "Numbering positions in HXB2": Korber et al—Journal unknown, affiliated with UCSD and referenced by LANL database. An in-depth annotation resource for identifying genes within HIV is also provided at www.hiv.lanl.gov/content/sequence/HIV/MAP/annotation.

It will be understood that the terms "based on" or "derived from" indicate that the nucleic acid sequence (or amino acid sequence) may comprise one or more modifications relative to a base sequence. Thus, the nucleic acid sequences may comprise one or more modifications (e.g., a deletion of one or more nucleotides, an addition of one or more nucleotides, a substitution of one or more nucleotides, or a combination thereof) relative to the corresponding wild-type nucleic acid sequence. In some embodiments, the nucleic acid sequences encoding the Pol polyprotein components and/or one or more accessory proteins have at least 60% identity (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%

identity) to the corresponding wild-type nucleic acid sequence. In some aspects, the wild-type Pol polyprotein is from the HIV-1 strain NL4-3.

The one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins may comprise one or more chemical or biological modifications relative to a naturally occurring nucleic acid molecule. The modifications may enhance nucleic acid or protein stability and/or transcriptional/translational efficiency and/or reduce immunogenicity. The modifications may include, but are not limited to, modified nucleobases, modified backbones (e.g., phosphoramidate, phosphorothioate, phosphorodithioate, O-methylphosphoramidite linkages and/or peptide nucleic acids).

In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins comprise one or more mutations that improve mRNA or corresponding mRNA stability. In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins comprise one or more mutations that improve Rev independence. In some embodiments, the nucleic acid sequences comprise mutations (e.g., a deletion of one or more nucleotides, an addition of one or more nucleotides, a substitution of one or more nucleotides, or a combination thereof) in one or more intrinsic instability (INS) elements. In some embodiments, a codon in an INS element is replaced with an alternate codon to preserve the protein sequence, while altering or removing the INS element. In some embodiments, the INS elements are replaced with codons optimized for expression in humans. In some embodiments, one or more INS elements are selected from TAGAT, ATAGA, AAAAG, ATAAA, and TTATA, or other INS elements (e.g., those described in Wolff et al. *Nucleic Acids Res.* 31(11):2839-51 (2003)). In some embodiments, INS elements are completely removed. In some embodiments, INS elements are partially removed. In some embodiments, the nucleic acid sequences are codon optimized for expression in a human. These modifications may be present in nucleotide sequences encoding one or more of reverse transcriptase, integrase, capsid, matrix, nucleocapsid, p6, Vif, Nef, Vpu, or Vpr.

In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins comprise one or more mutations that improve the stability of the corresponding protein. In some embodiments, a nucleic acid molecule of the disclosure encodes an integrase polypeptide comprising a stabilizing methionine-glycine dipeptide. In some embodiments, a nucleic acid of the disclosure encodes an integrase polypeptide comprising an N-terminal methionine-glycine dipeptide.

In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins comprise one or more mutations that improve translation efficiency in host cells. In some embodiments, the codons of the nucleic acid molecule of the disclosure are optimized for expression in a human.

In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins comprise one or more modifications selected from: codon optimization for expression in a host cell (e.g., human or other mammalian or avian cell), partial or complete removal of INS elements, and modification of integrase with a stabilizing methionine-glycine dipeptide. In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein com-

ponents and/or one or more accessory proteins are codon optimized for expression in a host cell (e.g., human or other mammalian or avian cell), comprise complete or partial removal of INS elements, and encode an integrase polypeptide modified with a methionine-glycine dipeptide.

In some embodiments, the one or more nucleic acid molecules encoding the Pol polyprotein components encode an integrase fused to a homing protein for site-directed genome integration (FIG. 10). A homing protein recognizes specific DNA sequences and thus directs the integrase to a specific sequence. This approach has the potential to be less genotoxic. In some embodiments, integrase is encoded by the Pol polyprotein. In some embodiments, the integrase is encoded as a separate unit. In some embodiments, the Pol polyprotein is fused to a homing protein. In some embodiments, a homing protein targets the genomic locus of the transgene. In some embodiments, a homing protein is an enzyme that recognizes a restriction site. Enzymes that recognize restriction sites are known in that art. See, e.g., www.neb.com/tools-and-resources/selection-charts/alpha-betized-list-of-recognition-specificities. Non-limiting

examples of enzymes that recognize restriction sites include, but are not limited to, AatII, AbaSI, Acc65I, AccI, AclI, AclI, AclI, AfeI, AflII, AflIII, AgeI, AhdI, AleI-v2, AluI, AlwI, AlwNI, ApaI, ApaLI, ApoI, AscI, AseI, AsiSI, AvaI, BsoBI, BsaI, BsaII, BsaHI, BsaJI, BsaWI, BsaXI, BseRI, BseYI, BsgI, BsiEI, BsiHKAI, BsiWI, BslI, BsmAI, BcoDI, BsmBI-v2, BsmFI, BsmI, BspCNI, BspEI, BspHI, BspI286I, BspMI, BfuAI, BsrBI, BsrDI, BsrFI-v2, BsrGI, BssHII, BssSI-v2, BstAPI, BstBI, BstEI, BstNI, BstUI, BstXI, BstYI, Bsu361, BtgI, BtgZI, BtsCI, BtsIMutI, BtsI-v2, Cac8I, ClaI, BspDI, CspCI, CviAII, CviKI-1, CviQI, DdeI, DpnI, DraI, DrdI, EaeI, EarI, EciI, Eco53kI, EcoNI, EcoO109I, EcoP15I, EcoRI, EcoRV, Esp3I, FatI, FaulI, Fnu4HI, FokI, FseI, FspEI, FspI, HaeII, HaeIII, HgaI, HhaI, HincII, HindIII, HinfI, HinfII, HpaI, HphI, HpyAV, HpyCH4III, HpyCH4IV, HpyCH4V, Hpy99I, Hpy188I, Hpy166I, Hpy188II, I-CeuI, I-SceI, KasI, KpnI, LpnPI, MboI, Sau3AI, DpnII, MboII, MfeI, MluCI, MluI, MlyI, MmeI, MnlI, MscI, MseI, MslI, MspAII, MspI, HpaII, MspJI, MwoI, NaeI, NarI, Nb.BbvCI, Nb.BsmI, Nb.BsrDI, Nb.BssSI, Nb.BtsI, NcoI, NdeI, NgoMIV, NlaIII, NlaIV, NmeAIII, NotI, NruI, NsiI, NspI, Nt.AlwI, Nt.BbvCI, Nt.BsmAI, Nt.BspQI, Nt.BstNBI, Nt.CviPII, PacI, PaqCI, PciI, PfiMI, PI-PspI, PI-SceI, PleI, PluTI, PmeI, PmlII, PpuMI, PshAI, PslI-v2, PspGI, PspOMI, PspXI, PstI, PvuI, PvuII, RsaI, RsrII, SacI, SacII, SalI, SapI BspQI, Sau96I, SbfI, ScrFI, SexAI, SfaNI, SfeI, SfiI, SfoI, SgrAI, SmaI, SmlI, SnaBI, SpeI, SphI, SrfI, SspI, StuI, StyD4I, SwaI, TaqI-v2, TfiI, TseI, ApeKI, Tsp45I, TspRI, Tth111I, PfiFI, XbaI, XcmI, XhoI, PaeR7I, XmaI, TspMI, XmnI, and ZraI. In some embodiments, a homing protein is a homing endonuclease. Homing endonucleases are known in the art, such as those disclosed in Stoddard, B. L., *Homing endonucleases from mobile group 1 introns: discovery to genome engineering*. Mobile DNA 5, 7 (2014). Non-limiting examples of homing endonucleases include I-TevI, I-HmuI, I-Bth035I, I-CreI, I-AniI, I-PpoI, and I-Ssp6803I. In some embodiments, the homing protein is I-PpoI. In some embodiments, the homing protein may be modified to remove any unwanted nuclease activity to prevent any genomic damage. In some embodiments, the homing protein is I-PpoI(N119A) or the like. In some embodiments, the integrase polypeptide

is modified to remove any unwanted nuclease activity. In some embodiments, the integrase polypeptide comprises a D64V mutation or the like.

In certain embodiments, the nucleic acid molecules encoding the Pol polypeptide components and/or one or more accessory proteins are RNA molecules (e.g., ssRNA molecules such as mRNA molecules). In some embodiments, the RNA molecule comprising a 5' UTR, a nucleic acid sequence encoding a retroviral Pol polypeptide, and a 3' UTR provides the RT and integrase functions. In some embodiments, a nucleic acid molecule comprising a 5' UTR, nucleic acid sequences encoding individual Pol polypeptide components, and a 3' UTR provides the RT and integrase functions. In some embodiments, all Pol polypeptide components are encoded on one RNA molecule with a 5' and 3' UTR. In some embodiments, the Pol polypeptide components are encoded on two or more RNA molecules each with a 5' and 3' UTR. In some embodiments, the Pol polypeptide components are encoded on one or more RNA molecules each with a 5' and 3' UTR and are separated by one or more polycistronic elements. The one or more polycistronic elements may be IRESes and/or 2A peptide-encoding sequences and/or other polycistronic elements. In some embodiments, the RNA molecules do not encode the Pol polypeptide protease, for example when the Pol polypeptide components for RT and integrase are encoded on separate RNA molecules or are separated by polycistronic elements such as intervening IRESes and/or 2A peptide-encoding sequences, and/or other polycistronic elements. In some embodiments, nucleic acid molecules encoding one or more accessory proteins to enhance reverse transcription and/or integration are additionally present. In some embodiments, the one or more accessory proteins are encoded in cis (i.e., encoded by the same RNA molecule) with one or more Pol polypeptide components. In some embodiments, the one or more accessory proteins are encoded in cis with all Pol polypeptide components. In some embodiments, the one or more accessory proteins are encoded in cis with the Pol polypeptide. Thus, in some embodiments, all Pol polypeptide components and accessory proteins are encoded on one RNA molecule with a 5' and 3' UTR. In some embodiments, the nucleotide sequences encoding the Pol polypeptide components and/or the accessory proteins are separated by one or more one or more polycistronic elements such as IRESes and/or 2A peptide-encoding sequences, and/or other polycistronic elements. In some embodiments, the one or more accessory proteins are encoded in trans (i.e., encoded by a separate nucleic acid molecule) from the Pol polypeptide components. In some embodiments, each of the Pol polypeptide components and accessory proteins are encoded by two or more (e.g., two, three, four, or five, or more) RNA molecules, each with a 5' and 3' UTR. In some embodiments, the UTRs include 3' or 5' sequences from mRNA molecules which are stable (including, but not limited to, globin, actin, GAPDH, tubulin, histone, or citric acid cycle enzymes).

In some embodiments, the nucleic acid molecules or compositions of the disclosure are capable of integrating the one or more transgenes into the host genome in the absence of functional retroviral Rev and/or Env proteins. In some embodiments, the compositions of the disclosure do not contain nucleic acid sequences that express proteins encoded by at least one of; retroviral rev gene and retroviral env gene. In some embodiments, the nucleic acid molecules or compositions of the disclosure do not contain nucleic acid sequences that express proteins encoded by both retroviral rev and env genes. In some embodiments, the compositions of the disclosure do not contain nucleic acid sequences

encoding the following retroviral Gag proteins: MA, SP1, SP2, and p6. In some embodiments, the compositions of the disclosure do not comprise any nucleic acid sequences encoding retroviral Gag proteins. In other embodiments, the compositions of the disclosure do not comprise nucleic acid sequences that are able to express all retroviral Gag proteins.

In some embodiments, the RNA molecules encoding retroviral Pol polypeptide components and/or one or more accessory proteins comprise one or more modified ribonucleosides. In some embodiments, the modified ribonucleoside is selected from 2-thiouridine, 5-azauridine, pseudouridine, 4-thiouridine, 5-methyluridine, 5-methylpseudouridine, 5-aminouridine, 5-aminopseudouridine, 5-hydroxyuridine, 5-hydroxypseudouridine, 5-methoxyuridine, 5-methoxypseudouridine, 5-ethoxyuridine, 5-ethoxypseudouridine, 5-hydroxymethyluridine, 5-hydroxymethylpseudouridine, 5-carboxyuridine, 5-carboxypseudouridine, 5-formyluridine, 5-formylpseudouridine, 5-methyl-5-azauridine, 5-amino-5-azauridine, 5-hydroxy-5-azauridine, 5-methylpseudouridine, 5-aminopseudouridine, 5-hydroxypseudouridine, 4-thio-5-azauridine, 4-thiopseudouridine, 4-thio-5-methyluridine, 4-thio-5-aminouridine, 4-thio-5-hydroxyuridine, 4-thio-5-methyl-5-azauridine, 4-thio-5-amino-5-azauridine, 4-thio-5-hydroxy-5-azauridine, 4-thio-5-methylpseudouridine, 4-thio-5-aminopseudouridine, 4-thio-5-hydroxypseudouridine, 2-thiocytidine, 5-azacytidine, pseudoisocytidine, N4-methylcytidine, N4-aminocytidine, N4-hydroxycytidine, 5-methylcytidine, 5-aminocytidine, 5-hydroxycytidine, 5-methoxycytidine, 5-ethoxycytidine, 5-hydroxymethylcytidine, 5-carboxycytidine, 5-formylcytidine, 5-methyl-5-azacytidine, 5-amino-5-azacytidine, 5-hydroxy-5-azacytidine, 5-methylpseudoisocytidine, 5-aminopseudoisocytidine, 5-hydroxypseudoisocytidine, N4-methyl-5-azacytidine, N4-methylpseudoisocytidine, 2-thio-5-azacytidine, 2-thiopseudoisocytidine, 2-thio-N4-methylcytidine, 2-thio-N4-aminocytidine, 2-thio-N4-hydroxycytidine, 2-thio-5-methylcytidine, 2-thio-5-aminocytidine, 2-thio-5-hydroxycytidine, 2-thio-5-methyl-5-azacytidine, 2-thio-5-amino-5-azacytidine, 2-thio-5-hydroxy-5-azacytidine, 2-thio-5-methylpseudoisocytidine, 2-thio-5-aminopseudoisocytidine, 2-thio-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-azacytidine, 2-thio-N4-methylpseudoisocytidine, N4-methyl-5-methylcytidine, N4-methyl-5-aminocytidine, N4-methyl-5-hydroxycytidine, N4-methyl-5-methyl-5-azacytidine, N4-methyl-5-amino-5-azacytidine, N4-methyl-5-hydroxy-5-azacytidine, N4-methyl-5-methylpseudoisocytidine, N4-methyl-5-aminopseudoisocytidine, N4-methyl-5-hydroxypseudoisocytidine, N4-amino-5-azacytidine, N4-aminopseudoisocytidine, N4-amino-5-methylcytidine, N4-amino-5-aminocytidine, N4-amino-5-hydroxycytidine, N4-amino-5-methyl-5-azacytidine, N4-amino-5-amino-5-azacytidine, N4-amino-5-hydroxy-5-azacytidine, N4-amino-5-methylpseudoisocytidine, N4-amino-5-aminopseudoisocytidine, N4-amino-5-hydroxypseudoisocytidine, N4-hydroxy-5-azacytidine, N4-hydroxypseudoisocytidine, N4-hydroxy-5-methylcytidine, N4-hydroxy-5-aminocytidine, N4-hydroxy-5-methyl-5-azacytidine, N4-hydroxy-5-amino-5-azacytidine, N4-hydroxy-5-hydroxy-5-azacytidine, N4-hydroxy-5-methylpseudoisocytidine, N4-hydroxy-5-aminopseudoisocytidine, N4-hydroxy-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-methylcytidine, 2-thio-N4-methyl-5-aminocytidine, 2-thio-N4-methyl-5-hydroxycytidine, 2-thio-N4-methyl-5-methyl-5-azacytidine, 2-thio-N4-methyl-5-amino-5-azacytidine, 2-thio-N4-methyl-5-hydroxy-5-azacytidine, 2-thio-N4-methyl-5-meth-

ylpseudoisocytidine, 2-thio-N4-methyl-5-aminopseudoisocytidine, 2-thio-N4-methyl-5-hydroxypseudoisocytidine, 2-thio-N4-amino-5-azacytidine, 2-thio-N4-aminopseudoisocytidine, 2-thio-N4-amino-5-methylcytidine, 2-thio-N4-amino-5-aminocytidine, 2-thio-N4-amino-5-hydroxycytidine, 2-thio-N4-amino-5-methyl-5-azacytidine, 2-thio-N4-amino-5-amino-5-azacytidine, 2-thio-N4-amino-5-hydroxy-5-azacytidine, 2-thio-N4-amino-5-methylpseudoisocytidine, 2-thio-N4-amino-5-aminopseudoisocytidine, 2-thio-N4-amino-5-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-azacytidine, 2-thio-N4-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-methylcytidine, N4-hydroxy-5-aminocytidine, 2-thio-N4-hydroxy-5-hydroxycytidine, 2-thio-N4-hydroxy-5-methyl-5-azacytidine, 2-thio-N4-hydroxy-5-amino-5-azacytidine, 2-thio-N4-hydroxy-5-hydroxypseudoisocytidine, N6-methyladenosine, N6-aminoadenosine, N6-hydroxyadenosine, 7-deazaadenosine, 8-azaadenosine, N6-methyl-7-deazaadenosine, N6-methyl-8-azaadenosine, 7-deaza-8-azaadenosine, N6-methyl-7-deaza-8-azaadenosine, N6-amino-7-deazaadenosine, N6-amino-8-azaadenosine, N6-amino-7-deaza-8-azaadenosine, N6-hydroxyadenosine, N6-hydroxy-7-deazaadenosine, N6-hydroxy-8-azaadenosine, N6-hydroxy-7-deaza-8-azaadenosine, 6-thioguanosine, 7-deazaguanosine, 8-azaguanosine, 6-thio-7-deazaguanosine, 6-thio-8-azaguanosine, 7-deaza-8-azaguanosine, 6-thio-7-deaza-8-azaguanosine, and N1-methylpseudouridine. In some embodiments, the modified ribonucleoside is pseudouridine or a derivative of pseudouridine. In some embodiments, the derivative of pseudouridine is N1-methylpseudouridine. In some embodiments, the modified ribonucleoside is N6-methyladenosine.

In some embodiments, the RNA molecules encoding retroviral Pol polyprotein components and/or one or more accessory proteins comprise one or more modifications including, but not limited to: a 5' terminal cap (e.g., a 5'-7mG cap structure); inclusion of a poly(rA) tail; alteration of the 3' UTR or the 5' UTR; complexing the mRNA with an agent (e.g., a protein or a complementary nucleic acid molecule); inclusion of elements which change the structure of an mRNA molecule (e.g., which form secondary structures); and reduction of the number of C and/or U residues. In some embodiments, the poly(rA) sequence is from about 10-500 nucleotides. In some embodiments, the poly(rA) sequence is at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 200, at least 300, at least 400, or at least 500 nucleotides. Other chemical and biological modifications to RNA molecules are known in the art.

Pol Polyprotein Component and Accessory Protein Nucleic Acid Templates

In one aspect, the disclosure also provides nucleic acid templates (e.g., a recombinant DNA construct) for the production of the nucleic acid molecules encoding the Pol polyprotein components and/or one or more accessory proteins described herein. In some embodiments, the nucleic acid templates are used to produce RNA molecules. In some embodiments, the RNA molecules are produced using cell-free methods (e.g., in vitro transcription or cell-free RNA synthesis). See, e.g., US Patent Publications No. 2017-0292138, 2018-0087045 2019-0144489, incorporated herein by reference in their entireties. In some embodiments, a nucleic acid template comprises a 5' UTR, a nucleic acid sequence encoding one or more Pol polyprotein compo-

nents, and a 3' UTR, in addition to RNA production elements (e.g., a T7 promoter, digestion sites, etc.). In some embodiments, a nucleic acid template comprises a T7 promoter, a 5' UTR, a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR, and optionally, one or more restriction digestion sites for template linearization. In some embodiments, a nucleic acid template comprises a T7 promoter, a 5' UTR, nucleic acid sequences encoding individual Pol polyprotein components, and a 3' UTR, and optionally, one or more restriction digestion sites for template linearization. In some embodiments, two or more nucleic acid templates, each with a T7 promoter, and a 5' and 3' UTR, encode the Pol polyprotein components. In some embodiments, one or more templates, each with a T7 promoter, and a 5' and 3' UTR, encode the Pol polyprotein components, wherein the Pol polyprotein components are separated by one or more polycistronic elements. The one or more polycistronic elements may be IRESes and/or 2A peptide-encoding sequences, and/or other polycistronic elements. In some embodiments, the nucleic acid templates do not encode the Pol polyprotein protease.

In some embodiments, the nucleic acid templates additionally comprise nucleic acid sequences encoding one or more accessory retroviral proteins to enhance reverse transcription and/or integration. In some embodiments, the one or more accessory proteins are encoded in cis (i.e., encoded by the same nucleic acid molecule) with one or more Pol polyprotein components. In some embodiments, the one or more accessory proteins are encoded in cis with all Pol polyprotein components. In some embodiments, the one or more accessory proteins are encoded in cis with the Pol polyprotein. In some embodiments, one nucleic acid template encodes all Pol polyprotein components and accessory proteins. In some embodiments, the nucleotide sequences encoding the Pol polyprotein components and/or the accessory proteins are separated by one or more polycistronic elements. The polycistronic elements may be IRESes and/or 2A peptide-encoding sequences, and/or other polycistronic elements. In some embodiments, the one or more accessory proteins are encoded in trans (i.e., encoded by a separate nucleic acid molecule) from the Pol polyprotein components. In some embodiments, each of the Pol polyprotein components and accessory proteins are encoded by two or more (e.g., two, three, four, or five, or more) nucleic acid templates, each with a T7 promoter, and a 5' and 3' UTR. In some embodiments, nucleic acid template comprises a T7 promoter, a 5' UTR, a nucleic acid sequence encoding gag-pol gene, a 3' UTR, and optionally, one or more restriction digestion sites for template linearization. In some embodiments, a nucleic acid template comprises a T7 promoter, a 5' UTR, a nucleic acid sequence encoding a gag-pol gene comprising a frameshift mutation, a 3' UTR, and optionally, one or more restriction digestion sites for template linearization. In some embodiments, the gag-pol gene does not encode the matrix protein.

In some embodiments, the UTRs include 3' or 5' sequences from mRNA molecules which are stable and highly translated (including, but not limited to, globin, actin, GAPDH, tubulin, histone, or citric acid cycle enzymes).

In some embodiments, the nucleic acid templates do not contain nucleic acid sequences that express proteins encoded by retroviral rev and env genes. In some embodiments, the nucleic acid templates do not comprise nucleic acid sequences encoding the following retroviral Gag proteins: MA, SP1, SP2, and p6. In some embodiments, the nucleic acid templates do not comprise nucleic acid sequences encoding MA. In some embodiments, the nucleic acid

templates comprise nucleic acid sequences encoding one or more of NC and CA. In some embodiments, the compositions of the disclosure do not comprise any nucleic acid sequences encoding retroviral Gag proteins (MA, CA, NC, SP1, SP2, P6). In other embodiments, the compositions of the disclosure do not comprise nucleic acid sequences that are able to express all retroviral Gag proteins. In some embodiments, the nucleic acid templates comprise nucleic acid sequences encoding one or more of NC, CA, MA, p6, Vif, Tat, Nef, Vpr, and Vpu. Exemplary nucleic acid templates encoding Pol polyproteins and/or accessory proteins are shown in FIGS. 4-5. In some embodiments, the RNA molecules are produced using cell-free methods (e.g., in vitro transcription, cell-free RNA synthesis).

Transgene-Encoding Nucleic Acid

In one aspect, the disclosure provides a nucleic acid molecule comprising one or more transgenes flanked by LTR sequences, which facilitate integration of the one or more transgenes into the host genome. The nucleic acid molecule may not comprise a viral packaging signal (e.g., the Psi packaging element) and/or a Rev response element (RRE). Thus, the transgene-encoding nucleic acid molecule encodes the one or more transgenes and contains the sequences that will assist in integrating the one or more transgenes into the host cell genome; however, the system does not allow for the production of any viral particles. In some embodiments, the transgene-encoding nucleic acid molecule comprises one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' LTR and a 3' LTR.

The term "long terminal repeat" or "LTR", as used herein, refers to the sequences of RNA or DNA that repeat hundreds or thousands of times and is found at either end of proviral DNA formed by reverse transcription of retroviral RNA. LTRs are used by viruses to integrate their genetic material into host genomes (e.g., the ends of the LTRs participate in integration of the provirus into the host genome). In some embodiments, the LTRs are self-inactivating. The 3' LTR may have a deletion (e.g., a deletion of the transcriptional enhancers or the enhancers and promoter in the U3 region of the 3' LTR) that is transferred to the 5' LTR after a single round of reverse transcription. The deletion in the 3' LTR (e.g., a deletion of the transcriptional enhancers or the enhancers and promoter in the U3 region of the 3' LTR) may be 10-50, 50-100, 100-150, 150-200, or more nucleotides. In some embodiments, the deletion in the 3' LTR is 100-150 nucleotides. In some embodiments, the deletion in the 3' LTR is 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, or 150 nucleotides. In some embodiments, the LTRs, Pol polyprotein components, and/or one or more accessory proteins are based on nucleic acid sequences from the same retrovirus. In some embodiments, the LTRs, the Pol polyprotein components and/or one or more accessory proteins are based on nucleic acid sequences from two or more different retroviruses. In some embodiments, the LTRs are based on LTRs from murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus

(AEV), human T-cell leukemia virus (HTLV), avian sarcoma leukosis virus (ASLV), endogenous xenotropic MuLV-related virus VP62 (XMRV), primate T-lymphotropic virus (PTLV), walleye dermal sarcoma virus (WDSV), the human foamy virus (HFV), Friend MLV (FrMLV), Avian sarcoma virus (ASV), Avian leukosis virus, Avian myeloblastosis virus, UR2 sarcoma virus, Y73 sarcoma virus, Jaagsiekte sheep retrovirus, Langur virus, Mason-pfizer monkey virus, Squirrel monkey retrovirus, Avian carcinoma mill hill virus 2, Bovine leukemia virus, Primate T-lymphotropic virus 1, Primate T-lymphotropic virus 2, Primate T-lymphotropic virus 3, Walleye dermal sarcoma virus, Walleye epidermal hyperplasia virus 1, Walleye epidermal hyperplasia virus 2, Chick syncytial virus, Feline leukemia virus, Finkel-Biskis-Jenkins murine sarcoma virus, Gardner-Amstein feline sarcoma virus, Gibbon ape leukemia virus, Guinea pig type-C oncovirus, Hardy-Zuckerman feline sarcoma virus, Harvey murine sarcoma virus, Kirsten murine sarcoma virus, Koala retrovirus, Moloney murine sarcoma virus, Porcine type-C oncovirus, Reticuloendotheliosis virus, Snyder-Theilen feline sarcoma virus, Trager duck spleen necrosis virus, Viper retrovirus, Wooly monkey sarcoma virus, Jembrana disease virus, Puma lentivirus, Bovine foamy virus, Equine foamy virus, Feline foamy virus, Brown greater galago prosimian foamy virus, Bornean orangutan simian foamy virus, Central chimpanzee simian foamy virus, Cynomolgus macaque simian foamy virus, Eastern chimpanzee simian foamy virus, Grivet simian foamy virus, Guenon simian foamy virus, Japanese macaque simian foamy virus, Rhesus macaque simian foamy virus, Spider monkey simian foamy virus, Squirrel monkey simian foamy virus, Taiwanese macaque simian foamy virus, Western chimpanzee simian foamy virus, Western lowland gorilla simian foamy virus, White-tufted-ear marmoset simian foamy virus, Yellow-breasted capuchin simian foamy virus, or a combination thereof. In some embodiments, the LTRs are based on LTRs from HFV. In some embodiments, the LTRs are lentiviral LTRs. In some embodiments, the lentiviral LTRs are based on LTRs from: human immunodeficiency virus (HIV) (e.g., HIV-1, HIV-2), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), jembrana disease virus, puma lentivirus, or a combination thereof. In some embodiments, the LTRs are based on LTRs from HIV-1 (e.g., HIV-1 with the genomic sequence in GenBank Accession No. AF033819). Other LTRs can be found in the art, for example on GenBank, as described above.

It will be understood that the terms "based on" or "derived from" indicate that the nucleic acid sequence may comprise one or more modifications relative to a base sequence. Thus, the LTRs may comprise one or more modifications (e.g., a deletion of one or more nucleotides, an addition of one or more nucleotides, a substitution of one or more nucleotides, or a combination thereof) relative to the corresponding wild-type nucleic acid sequence.

In some embodiments, the LTRs have at least 60% identity (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 9%, 96%, 95%, 98%, or 99% identity) to the wild-type LTRs.

HIV-1 RT depends on cytoplasmic tRNAs for priming of the viral genome prior to the reverse transcription reaction. In some embodiments, the transgene-encoding nucleic acid molecule may comprise one or more reverse transcriptase priming elements (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more).

In some embodiments, a reverse transcriptase priming element is a binding site for an endogenous tRNA. In some embodiments, the priming element is a binding site for lysyl tRNA. In some embodiments, a reverse transcriptase priming element is a synthetic primer binding site. In some embodiments, a reverse transcriptase priming element is a synthetic primer binding site, and one or more synthetic primers are provided for use in conjunction with the transgene-encoding nucleic acid molecule and the nucleic acid molecules encoding the Pol polyprotein components. A synthetic primer is a non-naturally occurring primer based on rational design. It will be understood that the synthetic primer may be based on naturally occurring sequences. A synthetic primer binding site is the complementary sequence. In some embodiments, a nucleic acid molecule of the disclosure is primed with one or more RNA or DNA priming oligonucleotides prior to delivery to target cells. In some embodiments, a nucleic acid molecule of the disclosure is co-delivered with one or more short RNA or DNA priming oligonucleotides. In some embodiments, an RNA or DNA priming oligonucleotide is 5-10, 10-15, 15-20, 20-25, 25-30, 5-15, 5-25, 5-30, 10-20, 10-25, 10-30, 15-25, or 15-30 nucleotides in length. In some embodiments, an RNA or DNA priming oligonucleotide is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In some embodiments, the RNA or DNA priming oligonucleotides may be 30-50 or 50-80 or 80-100, 100-150, or up to about 200 nucleotides. In some embodiments, the RNA or DNA priming oligonucleotides may be about 70-80 nucleotides. In some embodiments, the RNA or DNA priming oligonucleotides may be about 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, or 80 nucleotides. In some embodiments, the RNA or DNA priming oligonucleotides may be about 76 nucleotides. In some embodiments, the priming oligonucleotide is an engineered oligonucleotide that is complementary to the RT priming element. In some embodiments, the priming oligonucleotide is GUCCUGUUCGGGCGCCA (SEQ ID NO: 18) or GTCCCTGTTCGGGCGCCA (SEQ ID NO: 19).

Expression of the one or more transgenes may be controlled using one or more control sequences operably linked to the one or more transgenes, wherein the control sequences include promoters, enhancers, and other regulatory sequences. In some embodiments, the transfer nucleic acid comprises a single transgene linked to a promoter. In some embodiments, the transfer nucleic acid comprises two or more transgenes. In some embodiments, each transgene is operably linked to its own promoter. In some embodiments, a single promoter is operably linked to two or more transgenes. In some embodiments, some transgenes of the two or more transgenes are operably linked to their own promoters, and some transgenes of the two or more transgenes are operably linked to a common promoter.

The two or more transgenes may be separated by one or more polycistronic elements. The one or more polycistronic elements may be one or more IRESes and/or one or more 2A peptide-encoding sequences, and/or other polycistronic elements. An IRES is a nucleotide sequence that allows for the initiation of protein translation in the middle of a messenger RNA (mRNA) sequence. A 2A peptide is a small (18-22 amino acids) sequence that allows for efficient, stoichiometric production of discrete protein products within a single reading frame through a ribosomal skipping event within the 2A peptide sequence. Any other polycistronic element that will facilitate translation could also be used.

Any promoters functional in eukaryotic cells may be used in the present invention. In some embodiments, a promoter

may be one that is naturally associated with the transgene, and may be obtained by isolating the 5' non-coding sequence upstream of the coding segment and/or exon of a given gene. Such a promoter may be referred to as an endogenous promoter or a native promoter. In other embodiments, a transgene may be positioned under the control of a recombinant or heterologous promoter, which refers to a promoter that is not normally associated with the encoded nucleic acid sequence in its natural environment. In some embodiments, the promoter is tissue-specific or cell-specific (e.g., specific for bone marrow, hematopoietic stem cells (HSCs), T cells, liver, ocular tissue, or muscle, etc.). In some embodiments, the promoter is a constitutively active promoter. In some embodiments, the promoter is an inducible promoter. In some embodiments, the promoter may be a chimeric promoter comprising sequence elements from two or more different promoters. Suitable promoters include promoters derived from the genomes of viruses, such as polyoma virus, adenovirus, fowlpox virus, bovine papilloma virus, avian sarcoma virus, cytomegalovirus (CMV) (e.g., hCMV, mCMV), Rous sarcoma virus (RSV), and simian Virus 40 (SV40), or from heterologous mammalian promoters, such as the actin promoter, human elongation factor-1 alpha (EF1a) promoter, CAG promoter, thymidine kinase (TK) promoter, ubiquitin promoter, human phosphoglycerate kinase (PGK) promoter, human glyceraldehyde-3-phosphate dehydrogenase (GAPDH) promoter, or ribosomal protein promoter. Alternatively, tissue-specific promoters, such as rhodopsin (Rho) promoter, rhodopsin kinase (RhoK) promoter, cone-rod homeobox containing gene (CRX) promoter, neural retina-specific leucine zipper protein (NRL) promoter, Vitelliform Macular Dystrophy 2 (VMD2) promoter, tyrosine hydroxylase promoter, neuronal-specific enolase (NSE) promoter, astrocyte-specific glial fibrillary acidic protein (GFAP) promoter, human al-antitrypsin (hAAT) promoter, phosphoenolpyruvate carboxykinase (PEPCK) promoter, liver fatty acid binding protein promoter, Flt-1 promoter, IFN- β promoter (e.g., mIFN- β promoter), Mb promoter, SP-B promoter, SYN1 promoter, WASP promoter, SV40/hAlb promoter, SV40/CD43, SV40/CD45, NSE/RU5' promoter, ICAM-2 promoter, GPIIb promoter, GFAP promoter, fibronectin promoter, endoglin promoter, elastase-1 promoter, desmin promoter, B29 promoter, CD14 promoter, CD43 promoter, CD45 promoter, CD68 promoter, Nphs1 promoter, OG-2 promoter, and mNUS promoter may be used to drive transcription. In some embodiments, the promoter is a hCMV promoter.

Transcription of the one or more transgenes may be increased or modulated further by inserting one or more enhancer sequences into the transgene-encoding nucleic acid molecule. The enhancer may be positioned 5' or 3' to the promoter. Suitable enhancers may include an enhancer from a eukaryotic cell virus, such as the SV40 enhancer, the CMV early promoter enhancer, or the woodchuck hepatitis virus posttranscriptional regulatory element (WPRE). In some embodiments, the enhancer is a WPRE.

In some embodiments, a transgene-encoding nucleic acid molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, one or more promoters operably linked to one or more transgenes, and a 3' LTR. In some embodiments, the promoter is a hCMV promoter. In some embodiments, the transgene-encoding nucleic acid molecule further comprises one or more enhancers operably linked to the promoters and/or transgenes. In some embodiments, the enhancer is a WPRE element. In some embodiments, a transgene encoding nucleic acid molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, a

promoter operably linked to a transgene, an enhancer, and a 3' LTR. In some embodiments, a transgene-encoding nucleic acid molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, a hCMV promoter operably linked to a transgene, a WPRE enhancer, and a 3' LTR.

The transgene-encoding nucleic acid molecule may comprise one or more chemical or biological modifications relative to a naturally occurring nucleic acid molecule. The modifications may enhance stability and/or transcriptional/translational efficiency and/or reduce immunogenicity. The modifications may include, but are not limited to, modified nucleobases, modified backbones (e.g., phosphoramidite linkages and/or peptide nucleic acids).

In some embodiments, the transgene-encoding nucleic acid molecule is an RNA molecule (e.g., ssRNA molecule). In some embodiments, a transgene-encoding RNA molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, one or more promoters operably linked to one or more transgenes, and a 3' LTR. In some embodiments, the promoter is a hCMV promoter. In some embodiments, the transgene-encoding RNA further comprises one or more enhancers operably linked to the promoters and/or transgenes. In some embodiments, the enhancer is a WPRE element. In some embodiments, a transgene-encoding RNA molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, a promoter operably linked to a transgene, an enhancer, and a 3' LTR. In some embodiments, a transgene-encoding RNA molecule comprises a 5' LTR, one or more reverse transcriptase priming elements, a hCMV promoter operably linked to a transgene, a WPRE enhancer, and a 3' LTR.

In some embodiments, the transgene-encoding RNA molecule comprises one or more chemical or biological modifications relative to a naturally occurring RNA. In some embodiments, the transgene-encoding RNA molecule does not comprise any modified ribonucleosides. In some embodiments, the transgene-encoding RNA molecule comprises one or more modified ribonucleosides. Examples of suitable modified ribonucleosides are provided elsewhere in the description. In some embodiments, the RNA molecule comprises one or more modifications including but not limited to: modified ribonucleosides, a 5' terminal cap (e.g., a 5'-7mG cap structure); inclusion of a poly(rA) tail; complexing the mRNA with an agent (e.g., a protein or a complementary nucleic acid molecule); inclusion of elements which change the structure of an RNA molecule (e.g., which form secondary structures); and reduction of the number of C and/or U residues. In some embodiments, the poly(rA) sequence is from about 10-500 nucleotides. In some embodiments, the poly(rA) sequence is at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 200, at least 300, at least 400, or at least 500 nucleotides. In some embodiments, the transgene-encoding RNA molecule is not capped. Other chemical and biological modifications to RNA molecules are known in the art.

Transgene-Encoding Nucleic Acid Templates

In one aspect, the disclosure also provides nucleic acid templates (e.g., a recombinant DNA construct) for the production of the transgene-encoding nucleic acid molecule. In some embodiments, the nucleic acid template is used to produce a transgene-encoding RNA molecule. In some embodiments, the transgene-encoding RNA molecule is produced using cell-free methods (e.g., in vitro transcription or cell-free RNA synthesis). See, e.g., US Patent Publications No. 20170292138, 20180087045 20190144489, incor-

porated by reference herein in their entireties. In some embodiments, the nucleic acid template comprises a 5' LTR, one or more reverse transcriptase priming elements, one or more promoters operably linked to one or more transgenes, and a 3' LTR, in addition to RNA production elements (e.g., a T7 promoter, digestion sites, etc.). In some embodiments, a nucleic acid template comprises a 17 promoter, a 5' LTR, one or more reverse transcriptase priming elements, one or more promoters operably linked to one or more transgenes, and a 3' LTR, and optionally, one or more restriction digestion sites for template linearization. In some embodiments, the promoter is a hCMV promoter. In some embodiments, the nucleic acid template further comprises one or more enhancers operably linked to the promoters and/or transgenes. In some embodiments, the enhancer is a WPRE element. In some embodiments, a nucleic acid template comprises a T7 promoter, a 5' LTR, one or more reverse transcriptase priming elements, a promoter operably linked to a transgene, an enhancer, a 3' LTR and optionally, one or more restriction digestion sites for template linearization. In some embodiments, a transgene-encoding nucleic acid template comprises a 5' LTR, one or more reverse transcriptase priming elements, a hCMV promoter operably linked to a transgene, a WPRE enhancer, a 3' LTR and optionally, one or more restriction digestion sites for template linearization. An exemplary nucleic acid template encoding a transgene-encoding nucleic acid molecule is shown in FIG. 3.

Transgenes

In some embodiments, the transgene-encoding nucleic acid molecule encodes one or more transgenes. In some embodiments, the transgene-encoding nucleic acid molecule encodes two or more transgenes (e.g., 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more). In certain embodiments, the transgene-encoding nucleic acid molecule encodes only one transgene.

In some embodiments, a transgene encodes a therapeutic molecule, a diagnostic molecule, or reporter molecule.

In some embodiments, a transgene encodes a therapeutic molecule. The therapeutic molecule may be a nucleic acid or a polypeptide. Therapeutic molecules of the present disclosure may be used to, for example, replace a protein that is deficient or abnormal, augment an existing biological pathway, provide a novel function or activity, in a cell, tissue, organ, or subject. The therapeutic molecule may also be used to elicit an immune response. Exemplary types of transgenes include, but are not limited to, sequences encoding enzymes, co-factors, carrier proteins, transport proteins, cytokines, signaling proteins, suicide gene products, drug resistance proteins, tumor suppressor protein hormones, peptides with immunomodulatory properties, tolerogenic peptides, immunogenic peptides, antibodies and antigen-binding fragments thereof, anti-oxidant molecules, engineered immunoglobulin-like molecules, fusion proteins, immune co-stimulatory molecules, immunomodulatory molecules, chimeric antigen receptors, toxins, tumor suppressor proteins, growth factors, membrane proteins, receptors, vasoactive proteins, ligand proteins, antiviral proteins, ribozymes, RNAs, riboswitches, mRNA, RNA interference (e.g., shRNA, siRNA, microRNA) molecules, and derivatives thereof.

In some embodiments, therapeutic proteins encoded by the transgene include, but are not limited to, a cytokine, cystic fibrosis transmembrane conductance regulator protein (CFTR), dystrophin, utrophin, blood coagulation (clotting) factor (e.g., Factor XIII, Factor IX, Factor X, Factor VIII, Factor VIIa, protein C, Factor VII, B domain-deleted Factor VIII, or a high-activity or longer half-life variant of coagulation factor, or an active or inactive form of a coagulation factor), retinal pigment epithelium-specific 65 kDa protein

(RPE65), erythropoietin, LDL receptor, lipoprotein lipase, ornithine transcarbamylase, beta-globin, alpha-globin, spectrin, alpha-antitrypsin, adenosine deaminase (ADA), a metal transporter (ATP7A or ATP7), sulfamidase, an enzyme involved in lysosomal storage disease, hypoxanthine guanine phosphoribosyl transferase, beta-25 glucocerebrosidase, sphingomyelinase, lysosomal hexosaminidase, branched-chain keto acid dehydrogenase, insulin-like growth factor 1 or 2, platelet derived growth factor, epidermal growth factor, nerve growth factor, neurotrophic factor-3 and -4, brain-derived neurotrophic factor, glial derived growth factor, transforming growth factor alpha and beta, alpha-interferon, beta-interferon, interferon-gamma, interleukin-2, interleukin-4, interleukin-12, granulocyte-macrophage colony stimulating factor, lymphotoxin, herpes simplex virus thymidine kinase, cytosine deaminase, diphtheria toxin, cytochrome P450, deoxycytidine kinase, tumor necrosis factor, p53, Rb, Wt-1, NFL, Von Hippel-Lindau (VHL), SERCA2a, adenomatous polyposis coli (APC), VEGF, microdystrophin, minidystrophin, lysosomal acid lipase, arylsulfatase A and B, ATP7A and B, insulin, glucokinase, guanylate cyclase 2D (GUCY2D), Rab escort protein 1, LCA 5, ornithine ketoacid aminotransferase, Retinoschisis 1, USHIC, RP GTPase regulator (RPGR), MERTK, DFNBI, ACHM 2, 3 and 4, PKD-1 or PKD-2, TPPI, CLN2, a gene product implicated in lysosomal storage diseases (e.g., sulfatases, N-acetylglucosamine-1-phosphate transferase, cathepsin A, GM2-AP, NPC1, VPC2, a sphingolipid activator protein), and any other peptide or protein, or derivatives thereof that may have a therapeutic effect in a subject in need thereof.

It will be understood that the term "derivative thereof" includes any therapeutically or functionally active fragment, or modification of a base polypeptide. A derivative of a polypeptide may comprise one or more modifications (e.g., a deletion of one or more nucleotides, an addition of one or more nucleotides, a substitution of one or more nucleotides, or a combination thereof) relative to the corresponding wild-type polypeptide sequence. In some embodiments, a derivative is a homolog. Homology is generally inferred from sequence similarity between two or more nucleic acids or proteins (or sequences thereof). The precise percentage of similarity between sequences that is useful in establishing homology varies with the nucleic acid and protein at issue, but as little as 25% sequence similarity is routinely used to establish homology. Higher levels of sequence similarity, e.g., 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 99% or more can also be used to establish homology. In some embodiments, a derivative is a polypeptide that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to a base polypeptide. Methods for determining sequence similarity percentages are well known to those having ordinary skill in the art. In some embodiments, similarity can be determined using algorithms such as those described herein, including, for example, BLASTP and BLASTN algorithms, for example, using default parameters. In some aspects, the wild-type HIV-1 strain is NL4-3.

In some embodiments, the transgene encodes beta-globin or a derivative thereof. The beta-globin gene may be used for gene therapy to treat beta-thalassemia or sickle cell disease. In some embodiments, the beta-globin gene is a human beta-globin gene. In some embodiments, the transgene encoding beta-globin or a derivative thereof is operably linked to a hCMV promoter. In some embodiments, the

transgene encoding beta-globin or a derivative thereof is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for bone marrow or HSCs. In some embodiments, the transgene encoding beta-globin or a derivative thereof is operably linked to the native beta-globin promoter. In some embodiments, the transgene encoding beta-globin or a derivative thereof is further operably linked to a WPRE enhancer. In some embodiments, the transgene encodes a beta-globin polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_000509.1 or a fragment thereof. In some embodiments, the transgene encodes a beta-globin polypeptide having an amino acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NP_000509.1 or a fragment thereof. In some embodiments, the transgene comprises a beta-globin gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_000518.5 or a fragment thereof. In some embodiments, the transgene comprises a beta-globin gene having a nucleic acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NM_000518.5 or a fragment thereof.

In some embodiments, the transgene encodes CFTR or a derivative thereof. The CFTR gene may be used for gene therapy to treat cystic fibrosis. In some embodiments, the transgene encoding CFTR or a derivative thereof is operably linked to a hCMV promoter. In some embodiments, the transgene encoding CFTR or a derivative thereof is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for epithelial cells. In some embodiments, the transgene encoding CFTR or a derivative thereof is operably linked to the native CFTR promoter. In some embodiments, the transgene encoding CFTR or a derivative thereof is further operably linked to a WPRE enhancer. In some embodiments, the transgene encodes a CFTR polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_000483.3 or a fragment thereof. In some embodiments, the transgene encodes a CFTR polypeptide having an amino acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NP_000483.3 or a fragment thereof. In some embodiments, the transgene comprises a CFTR gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_000492.4 or a fragment thereof. In some embodiments, the transgene comprises a CFTR gene having a nucleic acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NM_000492.4 or a fragment thereof.

In some embodiments, the transgene encodes Factor VIII or a derivative thereof. The Factor VIII gene may be used for gene therapy to treat hemophilia B. In some embodiments, the Factor VIII gene is a human Factor VIII gene. In some embodiments, the transgene encoding Factor VIII or a derivative thereof is operably linked to a hCMV promoter. In

some embodiments, the transgene encoding Factor VIII or a derivative thereof is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for liver. In some embodiments, the promoter is specific for bone marrow or HSCs. In some embodiments, the transgene encoding Factor VIII or a derivative thereof is operably linked to the native Factor VIII promoter. In some embodiments, the transgene encoding Factor VIII or a derivative thereof is further operably linked to a WPRE enhancer. In some embodiments, the transgene encodes a Factor VIII polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_000123.1 or a fragment thereof. In some embodiments, the transgene encodes a Factor VIII polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_063916.1 or a fragment thereof. In some embodiments, the transgene encodes a Factor VIII polypeptide having an amino acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NP_000123.1 or NCBI Reference Sequence No. NP_063916.1 or a fragment thereof. In some embodiments, the transgene comprises a Factor VIII gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_000132.4 or a fragment thereof. In some embodiments, the transgene comprises a Factor VIII gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_019863.2 or a fragment thereof. In some embodiments, the transgene comprises a Factor VIII gene having a nucleic acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NM_000132.4 or NCBI Reference Sequence No. NM_019863.2, or a fragment thereof.

In some embodiments, the transgene encodes dystrophin or a derivative thereof. The dystrophin gene may be used for gene therapy to treat Duchenne Muscular Dystrophy (DMD). In some embodiments, the dystrophin gene is a micro-dystrophin or mini-dystrophin gene (Vincent et al., *Nature Genetics* 5:130 (1993); Wang et al., *Proc Natl Acad Sci USA* 97:13714-9 (2000) [mini-dystrophin]; Harper et al., *Nat Med* 8:253-61 (2002) [micro-dystrophin]). In some embodiments, the dystrophin gene is the human dystrophin gene. In some embodiments, the transgene encoding dystrophin or a derivative thereof is operably linked to a hCMV promoter. In some embodiments, the transgene encoding dystrophin or a derivative thereof is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for muscle cells. In some embodiments, the transgene encoding dystrophin or a derivative thereof is operably linked to the native dystrophin promoter. In some embodiments, the transgene encoding dystrophin or a derivative thereof is further operably linked to a WPRE enhancer. In some embodiments, the transgene encodes a dystrophin polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_000100.3 or a fragment thereof. In some embodiments, the transgene encodes a dystrophin polypeptide having an amino acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NP_000100.3

or a fragment thereof. In some embodiments, the transgene comprises a dystrophin gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_000109.4 or a fragment thereof. In some embodiments, the transgene comprises a dystrophin gene having a nucleic acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NM_000109.4 or a fragment thereof.

In some embodiments, the transgene encodes RP GTPase regulator (RPGR) or a derivative thereof. The RPGR gene may be used for gene therapy to retinitis pigmentosa. In some embodiments, the RPGR gene is the human RPGR gene. In some embodiments, the transgene encoding RPGR or a derivative thereof is operably linked to a hCMV promoter. In some embodiments, the transgene encoding RPGR or a derivative thereof is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for ocular tissue. In some embodiments, the transgene encoding RPGR or a derivative thereof is operably linked to the native RPGR promoter. In some embodiments, the transgene encoding RPGR or a derivative thereof is further operably linked to a WPRE enhancer. In some embodiments, the transgene encodes a RPGR polypeptide having the amino acid sequence corresponding to NCBI Reference Sequence No. NP_000319.1 or a fragment thereof. In some embodiments, the transgene encodes a RPGR polypeptide having an amino acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NP_000319.1 or a fragment thereof. In some embodiments, the transgene comprises a RPGR gene having the nucleic acid sequence corresponding to NCBI Reference Sequence No. NM_000328.3 or a fragment thereof. In some embodiments, the transgene comprises a RPGR gene having a nucleic acid sequence that is at least 60% identical (e.g., 70%, 80%, 90% to 100%, 95% to 100%, 98% to 100%, or 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical) to NCBI Reference Sequence No. NM_000328.3 or a fragment thereof.

In some embodiments, the transgene encodes a chimeric antigen receptor (CAR). The CAR may be used to treat cancer. In some embodiments, the transgene encoding the CAR is operably linked to a hCMV promoter. In some embodiments, the transgene encoding a CAR is operably linked to a tissue-specific or cell-specific promoter. In some embodiments, the promoter is specific for HSCs or T cells (e.g., cytotoxic T cells). In some embodiments, the transgene encoding the CAR is further operably linked to a WPRE enhancer.

In some embodiments, the transgene encodes a reporter molecule. The reporter molecule may be a nucleic acid or a polypeptide. A reporter molecule refers to a molecule that can be used to measure gene expression and generally produce a measurable signal such as fluorescence, luminescence, or color. In some embodiments, the polypeptide is luciferase. In some embodiments, the polypeptide is a fluorescent protein. Fluorescent proteins are known in the art, and are a subclass of fluorophores, which are fluorescent chemical compounds with the ability to re-emit light upon excitation. The fluorophore will absorb excitation light energy of a first specific wavelength, and then will re-emit

light energy at a second, longer specific wavelength. Each type of fluorophore responds to and emits differing wavelengths of light, depending on the nature of its chemical structure and environment. In some embodiments, the fluorescent protein useful in the invention includes, but is not limited to, green fluorescent protein (e.g., wt-GFP, EGFP, Emerald, Superfolder GFP, Azami Green, mWasabi, TagGFP, TurboGFP, AcGFP, ZsGreen, T-Sapphire, etc.), blue fluorescent protein, (e.g., EBFP, EBFP2, Azurite, mTagBFP, etc.), cyan fluorescent protein (e.g., ECFP, mECFP, Cerulean, mTurquoise, CyPet, AmCyan1, Midori-Ishi Cyan, TagCFP, mTFP1 (Teal), etc.), yellow fluorescent protein (e.g., EYFP, Topaz, Venus, mCitrine, YPet, TagYFP, PhiYFP, ZsYellow1, mBanana, etc.), orange fluorescent protein (e.g., Kusabira Orange, Kusabira Orange2, mOrange, mOrange2, dTomato, dTomato-Tandem, TagRFP, TagRFP-T, DsRed, DsRed2, DsRed-Express (TI), DsRed-Monomer, mTangerine, etc.), or red fluorescent protein (e.g., mRuby, mApple, mStrawberry, AsRed2, mRFP1, JRed, mCherry, HcRed1, mRaspberry, dKeima-Tandem, HcRed-Tandem, mPlum, AQ143, etc.). In some embodiments, the transgene encodes GFP. In some embodiments, reporter molecules can be fused in-frame to other protein coding sequences to identify where a protein is located in a cell, tissue, organ, or organism. Reporters for use in accordance with the present disclosure include any reporter described herein or known to one of ordinary skill in the art.

Methods of Use

The nucleic acids and compositions of the disclosure may be used to deliver any transgene with a biological effect to treat and/or ameliorate the symptoms associated with any disorder related to gene expression. Methods of the present disclosure may be used to, for example, replace a protein that is deficient or abnormal, augment an existing biological pathway, provide a novel function or activity, by integrating and stably expressing transgenes. Methods of the present disclosure may also be used to transiently express transgenes and elicit an immune response.

In one aspect, the disclosure provides a method for expressing a transgene in a subject in need thereof, the method comprising administering to the subject an effective amount of a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences, thereby expressing the one or more transgenes in the subject. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the disclosed transgene can be expressed in a subject in need thereof. The composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences can be used to express the one or more transgenes in the subject.

In one aspect, the disclosure provides a method for expressing a transgene in a cell, the method comprising delivering to the cell a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding

nucleic acid comprising one or more transgene sequences flanked by LTR sequences, thereby expressing the one or more transgenes in the cell. In some embodiments, the one or more nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. Suitable cell types include, but are not limited to, HSCs, liver cells, ocular cells (e.g., retinal cells), muscle cells, epithelial cells, T cells (e.g., cytotoxic T cells), etc. In some aspects, the disclosed transgene can be expressed in a cell. The composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences can be used to express one or more transgenes in the cell.

In one aspect, the disclosure provides a method for expressing a transgene in a tissue, the method comprising delivering to the tissue a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences, thereby expressing the one or more transgenes in the tissue. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. Suitable tissues include, but are not limited to, bone marrow, muscle, ocular tissue, cardiac tissue, liver tissue, epithelial tissue, connective tissue, nervous tissue, gastrointestinal tissue, etc. In some aspects, the disclosed transgene can be expressed in a tissue. The composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences can be used to express one or more transgenes in the tissue.

In one aspect, the disclosure provides a method for expressing a transgene in an organ, the method comprising delivering to the organ a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences, thereby expressing the one or more transgenes in the organ. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. Suitable organs include, but are not limited to, eyes, heart, lungs, liver, stomach, spleen, pancreas, small intestine, large intestine, kidneys, bone marrow, brain, etc. In some aspects, the disclosed transgene can be expressed in an organ. The composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising

ing one or more transgene sequences flanked by LTR sequences can be used to express one or more transgenes in the organ.

In one aspect, the disclosure provides a method of treating a condition using a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences, the method comprising administering the composition to a subject in need of treatment, thereby expressing the transgene in the subject. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the disclosed transgene can be used to treat a condition. The composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid and a transgene-encoding nucleic acid comprising one or more transgene sequences flanked by LTR sequences can be used to treat the condition.

In some embodiments, the one or more transgenes are integrated into the genome of a target cell (e.g., in a subject). In some embodiments, the one or more transgenes are stably expressed for at least a week, at least two weeks, at least a month, at least 6 months, at least a year, at least 2 years, at least 3 years, at least 4 years, or at least 5 years. In some embodiments, the one or more transgenes are stably expressed for 1-2 weeks, 2-4 weeks, 1-3 months, 3-6 months, 6-9 months, 9-12 months, 1-2 years, 2-3 years, 3-4 years, or 4-5 years, or longer. In some embodiments, the one or more transgenes are stably expressed for the lifetime of a subject. In some embodiments, the compositions of the disclosure are used to integrate one or more transgenes into the genome of a target cell (e.g., in a subject).

Any disease, disorder, or condition related to gene expression (e.g., where integration and stable expression of a transgene may be desired) may be treated. In some embodiments, the disease, disorder, or condition to be treated is a genetic condition. The genetic disease, disorder, or condition may be hereditary or non-hereditary. In some embodiments, the disease, disorder, or condition is a neurodegenerative, proliferative, inflammatory, or autoimmune disease, disorder, or condition. In some embodiments, the condition to be treated includes, but is not limited to, cystic fibrosis (and other diseases of the lung), hemophilia A, hemophilia B, beta-thalassemia, sickle cell disease, anemia and other blood coagulation disorders, Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, epilepsy and other neurological disorders, cancer, diabetes mellitus, muscular dystrophies (e.g., Duchenne, Becker), Leber congenital amaurosis, lysosomal storage disease, Usher's syndrome 1C, gyrate atrophy, connexin 26 deafness, achromatopsia, X-linked retinoschisis, polycystic kidney disease, Gaucher's disease, Hurler's disease, adenosine deaminase deficiency, glycogen storage diseases and other metabolic defects, Pompe's disease, congestive heart failure, retinitis pigmentosa, diseases of solid organs (e.g., brain, liver, kidney, heart), and the like. In some embodiments, the condition to be treated is a cancer. In some embodiments, the cancer is a blood cancer (e.g., lymphoma, leukemia, multiple myeloma, etc.), breast cancer, prostate cancer, cancer of the digestive system (e.g., esophageal cancer, stomach can-

cer, colorectal cancer), liver cancer, cervical cancer, ovarian or uterine cancer, pancreatic cancer, lung cancer, brain cancer (e.g., glioblastoma), skin cancer (e.g., melanoma), or sarcomas of muscle or nerve, etc.

In some embodiments, the disclosure provides a method of treating beta-thalassemia or sickle cell disease in a subject in need thereof comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding beta-globin, or a derivative thereof, flanked by LTR sequences. In some embodiments, the disclosure provides a method of treating sickle cell disease in a subject in need thereof comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding beta-globin, or a derivative thereof, flanked by LTR sequences. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of beta-thalassemia or sickle cell disease.

In some embodiments, the disclosure provides a method of treating cystic fibrosis in a subject in need thereof, comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding CFTR, or a derivative thereof, flanked by LTR sequences. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of cystic fibrosis.

In some embodiments, the disclosure provides a method of treating hemophilia B in a subject in need thereof comprising administering a composition one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding Factor VIII, or a derivative thereof, flanked by LTR sequences. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of hemophilia B.

In some embodiments, the disclosure provides a method of treating DMD in a subject in need thereof comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding dystrophin, or a derivative thereof, flanked by LTR sequences. In some embodiments, the nucleic acid molecules encoding the retroviral Pol poly-

protein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of DMD.

In some embodiments, the disclosure provides a method of treating retinitis pigmentosa in a subject in need thereof comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription/integration of a transgene-encoding nucleic acid, and a nucleic acid comprising a transgene encoding RPGR, or a derivative thereof, flanked by LTR sequences. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of retinitis pigmentosa.

In some embodiments, the disclosure provides a method of treating cancer in a subject in need thereof comprising administering a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components for reverse transcription and integration of a transgene-encoding nucleic acid, and a nucleic acid molecule comprising a transgene encoding a chimeric antigen receptor (CAR) flanked by LTR sequences. In some embodiments, the condition to be treated is a cancer. In some embodiments, the cancer is a blood cancer (e.g., lymphoma, leukemia, multiple myeloma, etc.), breast cancer, prostate cancer, cancer of the digestive system (e.g., esophageal cancer, stomach cancer, colorectal cancer), liver cancer, cervical cancer, ovarian or uterine cancer, pancreatic cancer, lung cancer, brain cancer (e.g., glioblastoma), skin cancer (e.g., melanoma), sarcomas of muscle or nerve, etc. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polyprotein components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some aspects, the compositions of the disclosure can be used in the treatment of cancer.

In some embodiments, the one or more transgenes are transiently expressed once integrated into the genome of a target cell (e.g., in a subject). The transient expression is sufficient to elicit an immune response. Any disease, disorder or condition which will benefit from an immune response can be treated. Thus, in some embodiments, the disclosure provides a method of eliciting an immune response in a subject in need thereof by transiently expressing one or more transgenes, the method comprising administering to the subject an effective amount of a composition comprising one or more nucleic acid molecules encoding retroviral Pol polyprotein components and a nucleic acid molecule comprising a transgene flanked by long terminal repeat sequences, thereby expressing the transgene in the subject. In some embodiments, the disease, disorder or condition is cancer and the transgene may encode a tumor antigen. In some embodiments, the disease, disorder or condition is an infectious disease (e.g., a disease caused by an infectious agent such as a pathogenic organism) and the transgene may encode an antigen associated with the infectious disease. The subject may be at risk for contracting an infectious disease and the composition may be administered

prophylactically. In some aspects, the compositions of the disclosure can be used to elicit an immune response.

Non-Viral Delivery Systems

The nucleic acid molecules of the disclosure are delivered to a cell, tissue, or subject via a non-viral delivery system. Any suitable non-viral delivery system known in the art may be employed to deliver the nucleic acid molecules of the disclosure to the target tissues and/or cells *ex vivo* or *in vivo*.

In some embodiments, the nucleic acid molecules (e.g., RNA, ssDNA, linear dsDNA) are packaged in lipid nanoparticles (LNPs). Suitable LNPs for packaging DNA molecules and RNA molecules (e.g., mRNA molecules) are known in the art. In some embodiments, the LNPs comprise one or more ionizable or cationic lipids (e.g., N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethylammonium chloride (DOTMA); 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP); 1,2-dioleoyl-3-dimethylammonium propane (DODAP); DODMA; Dlin-MC3-DMA (MC3); LP-01; diketopiperazine-based ionizable lipid, cKK-E12); one or more neutral phospholipids (e.g., cholesterol), one or more zwitterionic lipid molecules (e.g., 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE)), and/or one or PEG-modified lipids. In some embodiments, the LNPs comprise a polylactic acid (PLA) or poly(lactic-co-glycolic acid) (PGLA) polymers within a lipid monolayer. In some embodiments, ionizable lipids are pH-dependent ionizable materials.

In some embodiments, the non-viral delivery systems, including but not limited to, LNPs, are conjugated with polypeptides that specifically bind receptors on a target cell or tissue (e.g., viral envelope or capsid proteins, antibodies or antigen-binding fragments thereof, or other targeting moieties). In some embodiments, the LNPs are conjugated to a small molecule that facilitates targeting to a specific cell, tissue, or organ. Suitable cell types include, but are not limited to, HSCs, liver cells (e.g., hepatocytes), ocular cells (e.g., retinal cells), muscle cells, epithelial cells, T cells (e.g., cytotoxic T cells), etc. Suitable tissues include, but are not limited to, bone marrow, muscle, ocular tissue, cardiac tissue, liver tissue, epithelial tissue, connective tissue, nervous tissue, gastrointestinal tissue, etc. Suitable organs include, but are not limited to, eyes, heart, lungs, liver, stomach, spleen, pancreas, small intestine, large intestine, kidneys, etc. In some embodiments, the target cell or tissue include, but are not limited to, bone marrow, HSCs, epithelial cells, liver cells (e.g., hepatocytes), ocular cells (e.g., retinal cells), muscle cells, T cells (e.g., cytotoxic T cells).

In some embodiments, the nucleic acid molecules of the disclosure may be delivered naked, in an aqueous solution (such as a buffer, e.g., a sucrose citrate buffer), or in combination with lipids, polymers, peptides, or other compounds that facilitate entry into the cells. The nucleic acid molecules may be introduced into target cells or subjects using any suitable technique, e.g., by direct injection, micro-injection, transfection, nucleofection, electroporation, lipofection, high pressure spraying, biolistics, and the like. In some embodiments, the nucleic acid molecules are complexed with a cationic amphiphile. In some embodiments, the nucleic acid molecules are complexed with a transfection agent (e.g., Lipofectamine™, Lipofectin™, jetPEI, RNAiMAX, and InvivoFectamine, MegaFectin™, TransIT™). In some embodiments, the nucleic acid molecules are complexed with a cell-penetrating peptide (e.g., a polycationic or an amphipathic peptide).

In some embodiments, the nucleic acid molecules of the disclosure are packaged in liposomes, lipid nanoparticles, polypeptide nanoparticles, inorganic materials (e.g., silica

nanoparticles, gold nanoparticles, and inorganic-based carriers such as CaP), synthetic polymers, dendrimers, cationic nanoemulsions, polymeric nanoparticles, polymer and lipid hybrid carriers, or combinations thereof. In some embodiments, the nucleic acid molecules of the disclosure are packaged in viral capsids.

In some embodiments, the nucleic acid molecules of the disclosure are packaged in polymeric nanoparticles including, but not limited to, nanoparticles based on polyethyleneimine (PEI), polyacrylates, poly(β -amino esters) (PBAEs), and poly(aspartamides) (PAsp).

In some embodiments, the nucleic acid molecules of the disclosure are packaged in liposomes. Liposomes are artificially prepared vesicles which may primarily be composed of a lipid bilayer. Liposomes can be of different sizes such as, but not limited to, a multilamellar vesicle (MLV) which may be hundreds of nanometers in diameter and may contain a series of concentric bilayers separated by narrow aqueous compartments, a small unicellular vesicle (SUV) which may be smaller than 50 nm in diameter, and a large unilamellar vesicle (LUV) which may be between 50 and 500 nm in diameter. Liposome design may include, but is not limited to, opsonins or ligands in order to improve the attachment of liposomes to unhealthy tissue or to activate events such as, but not limited to, endocytosis. Liposomes may contain a low or a high pH in order to improve the delivery of the nucleic acid molecules. In some embodiments, the nucleic acid molecules of the disclosure are packaged in liposomes, such as those formed from 1,2-dioleoyloxy-N,N-dimethylaminopropane (DODMA) liposomes, DiLa2 liposomes from Marina Biotech (Bothell, Wash.), 1,2-dilinoleoyloxy-3-dimethylaminopropane (DLin-DMA), 2,2-dilinoleyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLin-KC2-DMA), and MC3 (US20100324120, the contents of which are herein incorporated by reference in its entirety). In some embodiments, the liposomes are PEGylated.

The nucleic acid molecules of the disclosure (e.g., packaged in liposomes, lipid nanoparticles, cationic nanoemulsions, polymeric nanoparticles, etc.) can be administered systemically or locally, intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostatically, intrapleurally, intratracheally, intranasally, intravitreally, intravaginally, intrarectally, topically, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularly, mucosally, intrapericardially, intraumbilically, intraocularly, orally, locally, by inhalation (e.g., aerosol inhalation), by injection, by infusion, by continuous infusion, by localized perfusion bathing target cells directly, via a catheter, via a lavage, in creams, or by any other method or any combination of the foregoing as would be known to one of ordinary skill in the art (see, for example, *Remington's Pharmaceutical Sciences* (1990), incorporated herein by reference).

In some embodiments, the nucleic acid molecules of the disclosure (e.g., packaged in liposomes, lipid nanoparticles, cationic nanoemulsions, polymeric nanoparticles) are administered in the form of a pharmaceutical composition comprising a pharmaceutically-acceptable carrier. As used herein, "pharmaceutically acceptable carrier" includes any and all solvents, dispersion media, coatings, surfactants, antioxidants, preservatives (e.g., antibacterial agents, antifungal agents), isotonic agents, absorption delaying agents, salts, preservatives, drugs, drug stabilizers (e.g., antioxidants), gels, binders, excipients, disintegration agents, lubricants, sweetening agents, flavoring agents, dyes, such like materials and combinations thereof, as would be known to one of ordinary skill in the art (see, for example, *Remington's*

ton's Pharmaceutical Sciences (1990), incorporated herein by reference). Except insofar as any conventional carrier is incompatible with the active ingredient, its use in the therapeutic or pharmaceutical compositions is contemplated. Compositions for lipid nanoparticles with biological active molecules and suitable carriers are disclosed in, for example, U.S. Pat. No. 7,404,969, which is incorporated herein by reference.

The nucleic acid molecules of the of the disclosure can be administered once, or alternatively they may be administered in a plurality of administrations. If administered multiple times, the compounds may be administered via different routes. For example, the first (or the first few) administrations may be made directly into the affected tissue while later administrations may be systemic.

Kits

In one aspect, the present disclosure provides kits for delivery of one or more transgenes. In some embodiments, the kit comprises a composition comprising an effective amount of one or more nucleic acid molecules encoding retroviral Pol polypeptide components, and a nucleic acid molecule encoding one or more transgenes flanked by LTR sequences according to the disclosure. In some embodiments, the kit comprises a composition comprising an effective amount of one or more nucleic acid molecules encoding retroviral Pol polypeptide components, and a nucleic acid molecule encoding one or more transgenes flanked by LTR sequences according to the disclosure, in unit dosage form. In some embodiments, the nucleic acid molecules encoding the retroviral Pol polypeptide components further encode one or more accessory retroviral proteins. In some embodiments, the composition comprises one or more additional nucleic acid molecules encoding one or more accessory retroviral proteins. In some embodiments, the kit comprises a composition comprising two nucleic acid molecules. In some embodiments, the kit comprises a composition comprising two or more nucleic acid molecules (e.g., 2, 3, 4, or 5). In some embodiments, the nucleic acid molecules are packaged or formulated for delivery in lipid nanoparticles. In some embodiments, the kit comprises a sterile container which comprises the composition. Such containers can be boxes, ampules, bottles, vials, tubes, bags, pouches, blister-packs, or other suitable container forms known in the art. Such containers can be made of plastic, glass, laminated paper, metal foil, or other materials suitable for holding medicaments. If desired, the composition is provided together with instructions for administering the cell to a subject in need thereof.

EXAMPLES

In order that the invention described herein may be more fully understood, the following examples are set forth. The Examples described in this Application are offered to illustrate the methods, compositions, and systems provided herein and are not to be construed in any way as limiting their scope.

Example 1: mRNA-Encoded Minimal Lentiviral Machinery for In Vivo Gene Therapy

A transgene-encoding template was designed as shown in FIG. 3. The transgene-encoding nucleic acid template comprises, from 5' to 3', a 5' LTR, a reverse transcriptase priming element, a promoter operably linked to a transgene, an enhancer, and a 3' LTR. Additional RNA production elements include a 5' T7 RNA polymerase promoter and a 3'

digest site. The RNA molecule produced from the transgene-encoding nucleic acid template comprises unmodified nucleosides since retroviral reverse transcriptases cannot copy modified ribonucleases (Swanstrom et al., *J. Biol. Chem.* 256:1115-1121(1981)). Additionally, the RNA has a 5'-7mG cap structure and a poly(rA) tail to enhance RNA stability.

Two minimal nucleic acid templates encoding Pol poly proteins (for reverse transcription/integration) were designed as shown in FIG. 4. The canonical gene construction provides for the gag and pol to be expressed inline, which requires translation slippage for Pol polyprotein expression. The novel design increases pol expression by processing pol from a separate construct as a polyprotein unit (or even potentially separate units as explained below). These novel designs encode different UTRs for increased mRNA stability and translation. This novel design optionally encodes accessory protein units essential for reverse transcription and integration with pol, which can be in cis with 2A. The first nucleic acid template comprises from 5' to 3', a 5' UTR, a nucleic acid sequence encoding retroviral Pol polyprotein components in the form of a Pol polyprotein, and a 3' UTR. Additional RNA production elements include a 5' T7 RNA polymerase promoter and a 3' digest site. The second nucleic acid template comprises from 5' to 3', a 5' UTR, a nucleic acid sequence encoding one or more retroviral accessory proteins, a nucleic acid sequence encoding retroviral Pol polyprotein components in the form of a Pol polyprotein, and a 3' UTR. The accessory protein units include retroviral elements not encoded by the pol polyprotein gene, and examples include, but are not limited, gag poly protein (p55), MA (p17), CA (p24), NC (p9), p6, tat, nef, vpr, vif, and vpu. Additional RNA production elements include a 5' T7 RNA polymerase promoter and a 3' digest site. The mRNA molecules produced from these nucleic acid templates comprise both modified (e.g., pseudouridine) and unmodified nucleosides. The use of modified nucleosides reduces the immunogenicity of the final mRNA molecule. Additionally, the mRNA has a 5'-7mG cap structure and a poly(rA) tail to ensure efficient translation and stability. For example, the 3' UTR could encode poly A tail.

At least three different constructs for expressing the multiple-functional units of the Pol polyprotein were designed as shown in FIG. 5. In the first, as shown in FIG. 5A, the construct provides the protease (PR), reverse transcriptase/RNase H (RT), and integrase (IN) as a canonical unit, where each component is separated by autoprocessing by PR. In the second, as shown in FIG. 5B, the Pol polyprotein component individual units RT and IN are encoded on a bicistronic construct. In the third, as shown in FIG. 5C, the Pol polyprotein component individual units are encoded on separate constructs. There are various scenarios in which these constructs could be used. First, all of the Pol polyprotein components can be encoded as a polyprotein on one mRNA with a 5' and 3' UTR (represented by FIG. 5A), wherein the PR protease is required when encoding the components as polyprotein (pol). Second, the Pol polyprotein components RT and IN units can be encoded on one or more constructs each with a 5' and 3' UTR with intervening IRES or 2A sequences (represented by FIG. 5B), wherein the PR protease is not required when encoding the elements as "non-polyprotein" individual units. Third, the Pol polyprotein components RT and IN units can be encoded on two or more constructs each with a 5' and 3' UTR (represented by FIG. 5C). Fourth, the Pol polyprotein components RT and IN units and accessory protein units can be encoded on one construct each with a 5' and 3' UTR with intervening IRES

or 2A sequences. In this scenario, when the accessory elements are encoded as a polyprotein, for example the Gag polyprotein, the PR protease will be required. And as a fifth, non-limiting scenario, the Pol polyprotein components RT and IN units and accessory protein units can be encoded on two or more constructs each with a 5' and 3' UTR.

Example 2: In Vivo Delivery of Beta-Globin to Treat Sickle Cell Disease

A transgene-encoding nucleic acid templates as described in Example 1 comprising a beta-globin transgene is used to produce an RNA molecule comprising the beta-globin gene flanked by LTRs for integration into a host cell genome. A nucleic acid template comprising, from 5' to 3', a 5' UTR, a nucleic acid sequence encoding retroviral Pol polyprotein components in the form of a Pol polyprotein, and a 3' UTR, as described in Example 1 is used to produce an RNA molecule encoding retroviral Pol proteins.

An effective amount of a lipid nanoparticle composition comprising the two RNA molecules is administered to a patient suffering from sickle cell disease to treat sickle cell disease.

Example 3: Improving RNA Stability and Translation in Human Cells

The lentivirus-derived genes forming the basis of the RNA-mediated transgene integration systems described herein can pose challenges for synthetic gene expression in cells. Lentiviral replication steps undergo tightly-controlled spatiotemporal processes that are orchestrated by a full complement of viral genes and regulatory elements, structural proteins, replicative enzymes, and interaction with host factors critical for replication. This example describes several modifications to increase stability and efficiency of the RNA-based integration system.

Increased Stability of Functional Unit RNAs

The one or more nucleic acid molecules encoding one or more Pol polyprotein components and/or accessory proteins are also referred to as "functional unit nucleic acids." The generation of a minimal, integration-competent RNA system generally entails the removal of viral genes that do not directly influence reverse transcription and proviral integration. However, expression of wild-type individual viral RNAs (i.e., Reverse Transcriptase (RT) and integrase (IN)) in the absence of the full-length viral genome is challenging without accessory proteins, such as for example Rev. In addition, wild-type HIV-1 RNAs are inherently unstable due to the presence of intrinsic instability elements (INS), short, 5-nucleotide repeats present throughout transcripts (Wolff et al. *Nucleic Acids Res.* 31(11):2839-51 (2003)). During HIV-1 infection, Rev can counteract the effect of INS because it can bind to and stabilize the RNA transcripts. To increase RNA stability in the absence of Rev, five major INS sequences were identified, removed, and replaced throughout functional unit mRNAs: TAGAT, ATAGA, AAAAG, ATAAA, TTATA or other INS elements (e.g., such as those described for example in Wolff et al. *Nucleic Acids Res.* 31(11):2839-51 (2003)). INS removal can be partial or complete. The INS sequences can be replaced with alternate codons, preserving the protein sequence while increasing the stability of the RNA.

Furthermore, human codon optimization (hCO) results in increased gene expression in human cells. Previous studies using full-length gag-pol genes demonstrated Rev-independence following human codon optimization (Kotsopoulou et

al. *J Virol.* 74(10): 4839-52 (2000)). hCO functional unit mRNAs were generated for increased translation and stability in target cells. In another iteration, remaining INS elements were removed and replaced from hCO sequences.

These improvements have been applied to all Pol polypeptide components and accessory proteins, including RT, IN, Matrix (MA), Capsid (CA), Nucleocapsid (NC) and Vpr RNAs.

Increased Stability of Functional Unit Proteins

Due to their cytoplasmic translation from input mRNAs, the functional units (e.g., Pol polypeptide components and accessory proteins) in this system can lack the spatiotemporal coordination observed in incoming virus particles. Unlike incoming virus particles, in which active IN and RT are packaged in close proximity to viral genomes, this RNA-based system first necessitates translation of incoming functional unit mRNAs and subsequent interaction of newly-translated functional units with the transgene-encoding RNA. Thus, protein lifetime of the functional units is important.

Wild-type IN protein is short-lived in cells, as the N-terminal residues of the wild-type IN are substrates for proteolytic degradation (Mulder et al. *J Biol Chem.* 275(38): 29749-53 (2000)). Previous studies using expression plasmids demonstrated that this degradation is counteracted by the addition of an N-terminal Methionine-Glycine dipeptide (Cherepanov et al. *FASEB J.* 14(10):1389-99 (2000)). Several N-terminal Met-Gly IN constructs were generated. These mRNAs are derived from WT HIV genes, are hCO, and/or have undergone removal and replacement of INS constructs.

Cytoplasmic Redirection of Pol-Derived Functional Units in Target Cells

In some embodiments, RNAs encoding singular gag or pol genes were generated, which comprise critical structural proteins necessary for nuclear pore crossing during the pre-integration replication step, or RT and IN, respectively. This strategy is advantageous in that the functional unit RNA sequences are translated as natural Gag or Pol polypeptides, while separation of the gag and pol genes enables stoichiometric optimization that favors intracellular activity. Expression of gag and pol genes in wild-type ratios results in assembly, budding, and release of new viruses from the producer cell into the extracellular space (Sundquist et al. *Cold Spring Harb Perspect Med.* 2(7):a006924 (2012)). While there is some evidence for functional RT and IN activity in the producer cell (Al Mosabbir et al. *Biotechnol. Lett.* 38(10):1715-21 (2016)), expression of new viral gene products in the producer cell is primarily understood to favor new virion formation. Several studies of expression plasmids encoding gag and pol have indicated redirection of viral gene expression away from particle formation toward intracellular activity of pol products (Al Mosabbir et al. *Biotechnol. Lett.* 38(10):1715-21 (2016), Karacostas V et al. *Virology* 193(2):661-71 (1993)). To this end, several mechanisms involving DNA plasmid transfection have been established: (1) deletion of the gag sequence encoding matrix protein, which anchors the Gag polypeptide to the plasma membrane during viral budding (Al Mosabbir et al. *Biotechnol. Lett.* 38(10):1715-21 (2016)), and (2) a single-nucleotide insertion disrupting the gag-pol frameshift mechanism (Karacostas V et al. *Virology* 193(2):661-71 (1993)). While these modifications have been described in the context of full-length viral genomes, they have not been explored for (1) extra-genomic singular gene expression and (2) synthetic, functional RNA. To facilitate intracellular activity of functional units expressed as gag- and pol-

derived polypeptides, several novel full-length gag and pol RNAs were engineered: (1) Matrix-deleted gag-pol RNAs for prevention of particle budding, (2) Matrix-deleted singular gag RNAs (3) hCO matrix-deleted gag-pol RNAs for prevention of particle budding (4) RNAs containing mutations that result in gag and pol in a singular reading frame, (5) hCO gag-pol frameshifted RNAs for intracellular pol activity, (6) transfection of singular gag or pol RNAs in optimized ratios that result in intracellular Pol product accumulation.

Primed RNA Genomes for Facilitation of the Reverse Transcription Reaction

HIV-1 RT depends on cytoplasmic tRNAs for priming of the viral genome prior to the reverse transcription reaction. Increased efficiency of the reverse transcription reaction in vitro through priming the viral genome with short RNA or DNA oligos has been observed (Iwatani et al. *J Biol Chem.* 278(16):14185-95 (2003)). As a strategy for increasing the efficiency of the reverse transcription reaction in our integration-competent system, transgene-encoding RNAs are primed with short RNA or DNA oligos prior to their co-delivery into target cells with functional unit mRNA(s).

Uncapped Transgene-Encoding RNA

HIV-1 RNA genomes are capped following transcription in producer cells. Following viral entry, viral genomes are decapped by cellular host factors, such as RNA debranching enzyme 1 (DBR1), prior to reverse transcription (Galvis et al. *J Virol.* 2017 November 14; 91(23): e01377-17 (2017)). To increase efficiency of the reverse transcription step, uncapped transgene-encoding RNAs were produced and cotransfected with functional unit mRNA(s).

Example 4: Materials and Methods

The materials and methods used in the studies described in Examples 5 and 6 are described.

IVT Template Design

DNA template sequences for IVT of functional unit RNA, including UTRs and 5' T7 promoters, were cloned into the pTWIST Amp cloning vector (Twist Bioscience). IVT template sequences for the transgene-encoding RNA, including LTRs and 5' T7 promoters, were cloned into the pMK backbone from GeneArt (ThermoFisher).

RNA Manufacturing and Purification

Plasmids containing IVT templates were linearized by digestion with BsmBI, or Bsu36I and SfiI (New England Biolabs). Digested DNA was isopropanol precipitated. Digested DNA template (30 ng/μL) was incubated for 4 h at 37° C. in the presence of NTPs (transgene-encoding RNA: 7.5 mM CTP, 1.5 mM GTP, 7.5 mM UTP; functional unit RNA: 4 mM ATP, 4 mM CTP, 4 mM GTP, 4 mM UTP), CleanCap (transgene-encoding RNA: CleanCap GG, 6 mM; functional unit RNA: CleanCap AG, 5 mM), 0.1M MgSO₄, 0.1M Spermidine, T7 RNA Polymerase (transgene-encoding RNA: 0.2 mg/mL; Functional Unit RNA: 0.1 mg/mL), and 40 U/mL TIPP (New England Biolabs). Reactions were treated with DNase (Thermo Fisher) and incubated for 15 minutes at 37° C. RNA was LiCl precipitated and resuspended in THE RNA storage solution (Thermo Fisher). In vitro transcribed RNA was enzymatically polyadenylated using *E. coli* Poly(A) polymerase (New England Biolabs). RNA (0.25 μg/μL) was incubated at 37° C. for 20 min in the presence of *E. coli* poly(A) polymerase (0.1 U/μL), 1× PolyA polymerase reaction buffer, and 10 mM ATP. RNA was LiCl precipitated and resuspended in THE RNA storage solution (Functional Unit RNA) or purified by HPLC (trans-

gene-encoding RNA). Polyadenylation was confirmed using a BioAnalyzer (Agilent Technologies).

RNA Transfection

293FT cells (Thermo Fisher) were transfected with RNA using Lipofectamine 2000 (Thermo Fisher). For transgene expression experiments, cells seeded in 48-well plates were transfected with 500 ng of total RNA in the indicated ratios, complexed with 0.5 μ L of Lipofectamine 2000 per well. For Western blotting, 293FT cells seeded in 12-well plates were transfected with 1 μ g of integrase RNA and 2 μ L Lipofectamine 2000 per well. For confocal microscopy, 293FT cells were seeded on Poly-D-Lysine-coated #1 coverslips (Electron Microscopy Sciences) in 24-well plates. Cells were transfected with 1 μ g of the integrase RNA per well of a 24-well plate.

Western Blotting

48 h post-transfection, transfected cells were resuspended in RIPA buffer (Pierce, Thermo Fisher) containing Halt protease inhibitor (Pierce, Thermo Fisher). Following two freeze-thaw cycles at -80° C., cell suspensions were lysed in the presence of Benzoase nuclease at 4° C. (Sigma Aldrich). Lysates were incubated at 37° C. for 5 minutes, treated with EDTA, and clarified via centrifugation at $21,000\times g$ for 20 minutes at 4° C. Lysates were electrophoresed on NuPage 4-12% Bis-Tris gels (Thermo Fisher) and transferred to 0.2 μ m PVDF membranes (Trans-Blot Turbo Mini, Bio-Rad). Membranes were probed with primary antibodies to integrase (Abcam, [In-2], ab66645) and β -Actin (Abcam, ab8227). Integrase and β -Actin were detected with fluorescent secondary goat anti-mouse and goat anti-rabbit antibodies (Abcam), respectively. Blots were imaged on a Li-Cor imager.

Immunocytochemistry and Confocal Microscopy

24 h following transfection, transfected 293FT cells were washed with PBS and fixed for 7 min at room temperature in 4% paraformaldehyde (Electron Microscopy Sciences). After fixation, cells were washed with PBS. Fixed cells were stained overnight with 300 nM DAPI (Biotium), ViaFluor-647 (Biotium), and 1:100 anti-integrase antibody (Abcam, [In-2], ab66645) conjugated with CF-568 dye (Mixer-and-Stain, Biotium) in blocking buffer (5% FBS, 1% BSA, and 1 $\times$ TBS Tween (Thermo Fisher) in PBS) overnight at room temperature. Stained coverslips were washed with PBS and mounted onto Superfrost Plus microscope slides (Fisher) using ProLong Glass (ThermoFisher). Samples were visualized using a Nikon/Yokogawa CSU-W1 spinning disk confocal microscope.

Nanoluciferase Assays

Cells were lysed for 10 minutes at 37° C. in Nano-Glo Buffer (Promega) without furimazine. Nanoluciferase assays were conducted in 96-well opaque plates (Costar) via the addition of 25 μ L of Nano-Glo buffer with 1:50 furimazine (Promega) to 20 μ L of cell lysate per well. Lumi-

nescence was read on a BioTek HTX plate reader immediately after furimazine addition.

Example 5: RNA-Derived Functional Units Drive Reporter Gene Expression from Transgene-Encoding RNAs

293 FT cells were transfected with an RNA encoding the Gag polypeptide, an RNA encoding the Pol polypeptide, and a transgene-encoding RNA encoding nanoluciferase (Nluc), or with the transgene-encoding RNA alone. The RNA constructs are shown in FIG. 7A. Cells were harvested for Nluc assay at 3 days post-transfection. Nluc activity was assessed via a luminometer. Nluc activity is displayed as light units relative to background (RLU) (FIG. 7B). The ratio of RNA species in 500 ng RNA transfection per well of a 48-well plate is indicated. Statistical significance was determined and confirmed by Student's t-test; $n=3$. These results demonstrate that lentiviral RNA-derived functional units are able to drive reporter gene expression from a transgene-encoding RNA flanked by LTRs. The results further guide stoichiometries for individual mRNA functional unit plus transgene-encoding RNA transfections.

Example 6: Stabilized Functional Unit Expression

Optimized integrase (IN)-encoding functional unit mRNAs with one or more of the modifications described in Example 3, to increase RNA and/or protein stability, are more stably expressed than wild-type integrase and localize to the nucleus in human cells. RNA construct design elements of RNAs encoding wild-type IN, stabilized Met-Gly, hCO IN, and stabilized Met-Gly hCO Δ INS IN are shown in FIG. 8A. In Δ INS IN, instability sequences (INS) were identified and codon-optimized to increase RNA stability.

293FT cells were transfected with each indicated RNA and harvested for western blot lysis at 48 h post-transfection (p.t.). Integrase was expressed at higher levels from the optimized constructs compared to the construct encoding the wild-type IN. Wild-type IN protein bands were undetectable at 48 h p.t., in contrast to those from optimized constructs (FIG. 8C). Subcellular localization of optimized IN (Met-Gly hCO Δ INS IN) following RNA transfection into 293FT cells was also determined (FIG. 8D). Cells were transfected with optimized IN RNA fixed 24 h post-transfection, and processed for immunofluorescence. Fixed cells were stained with conjugated anti-N antibody. DAPI (nuclei), and phalloidin-647 (actin). Integrase was strongly expressed and localized to the nucleus. These results demonstrate that optimized integrase-encoding functional unit mRNAs demonstrated enhanced expression and localization to target organelles.

A summary of exemplary nucleotide sequences described in the examples is provided below in Table 1.

TABLE 1

Exemplary nucleotide sequences		
Description	Relevant FIG.	SEQ ID NO.
Transgene-encoding nucleic acid molecule with GFP transgene	3	1
Transgene-encoding nucleic acid molecule with mEYFP transgene	3	2
Transgene-encoding nucleic acid molecule with smURFP transgene	3	3
Transgene-encoding nucleic acid molecule with CyRFP transgene	3	4
Transgene-encoding nucleic acid molecule with firefly luciferase transgene	3	5
Canonical gag-pol	4	6

TABLE 1-continued

Exemplary nucleotide sequences		
Description	Relevant FIG.	SEQ ID NO.
HIV-1 gag	7	7
Human codon optimized Met-Gly, IMS-deleted IN-I-PpoI N119A	10	8
Human codon optimized Pol-I-PpoI N119A	10	9
Transgene-encoding nucleic acid molecule with nanoluciferase transgene	3, 7	10
Human codon optimized pol RNA construct derived from pSYNGP plasmid with methionine-glycine dipeptide	4, 5	11
HIV-1 pol	4, 5, 7	12
Human codon optimized IN with Met-Gly dipeptide	5, 7	13
Human codon optimized IN with instability sequences replaced, methionine glycine dipeptide	5, 7	14
HIV-1 IN	5, 8	15
Human codon optimized RTp51	5, 8	16
Human codon optimized RTp66	5, 8	17

EMBODIMENTS

The following embodiments are within the scope of the present disclosure. Furthermore, the disclosure encompasses all variations, combinations, and permutations of these embodiments in which one or more limitations, elements, clauses, and descriptive terms from one or more the listed embodiments is introduced into another listed embodiment in this section. For example, any listed embodiment that is dependent on another embodiment can be modified to include one or more limitations found in any other listed embodiment in this section that is dependent on the same base embodiment. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should be understood that, in general, where the disclosure, or aspects of the disclosure, is/are referred to as comprising particular elements and/or features, certain embodiments of the invention or aspects of the invention consist, or consist essentially of, such elements and/or features. It is also noted that the terms “comprising” and “containing” are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

1. A composition comprising:

- (i) one or more nucleic acid molecules encoding one or more Pol polyprotein components flanked by 5' and 3' untranslated regions (UTRs); and
- (ii) a nucleic acid molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR,

wherein the composition does not contain nucleic acid sequences that express proteins encoded by retroviral rev and env genes.

2. The composition of embodiment 1, wherein the expression of the Pol polyprotein components do not require translational slippage from an inline gag gene.

3. The compositions of embodiments 1-2, wherein the one or more nucleic acid molecules of (i) do not encode the Gag polyprotein.

4. The compositions of embodiments 1-3, wherein the one or more nucleic acid molecules of (i) is a nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding a Pol polyprotein, and a 3' UTR.

5. The compositions of embodiments 1-3, wherein the one or more nucleic acid molecules of (i) is a nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein components reverse transcriptase and integrase, and a 3' UTR.

6. The compositions of embodiment 5, wherein the Pol polyprotein components reverse transcriptase and integrase are expressed on a bicistronic construct.

7. The compositions of embodiments 5-6, wherein the Pol polyprotein components reverse transcriptase and integrase are expressed with an intervening internal ribosome entry sites (IRESes) or 2A peptide-encoding sequence.

8. The compositions of embodiments 1-3, wherein the one or more nucleic acid molecules of (i) is two nucleic acid molecules, wherein one of the two nucleic acid molecules comprises a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein component reverse transcriptase, and a 3' UTR, and wherein the second of the two nucleic acid molecules comprises a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein component integrase, and a 3' UTR.

9. The compositions of embodiments 1-8, wherein the one or more nucleic acid molecules of (i) also comprises a nucleic acid encoding one or more Gag polyprotein accessory proteins.

10. The composition of embodiment 9, wherein the one or more Gag polyprotein accessory proteins are encoded on the same nucleic acid molecule as the Pol polyprotein components.

11. The composition of embodiment 10, wherein the one or more Gag polyprotein accessory proteins are expressed with an intervening internal ribosome entry sites (IRESes) or 2A peptide-encoding sequence.

12. The composition of embodiment 9, wherein the one or more Gag polyprotein accessory proteins are encoded by one or more different nucleic acid molecules as the Pol polyprotein components, wherein each nucleic acid molecule comprises a 5' UTR and a 3' UTR.

13. The composition of any one of embodiments 9-12, wherein the Gag polyprotein accessory proteins are selected from nucleocapsid (NC), capsid protein (CA), matrix protein (MA), p6, viral infectivity factor (Vif), transactivator-of

transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu).

14. The composition of embodiment 13, wherein the Gag polyprotein accessory proteins comprise nucleocapsid (NC), capsid protein (CA), matrix protein (MA), p6, viral infectivity factor (Vif), transactivator-of transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu).

15. The composition of embodiment 14, wherein the Gag polyprotein accessory proteins are encoded by the Gag polyprotein.

16. The composition of any one of embodiments 1-15, wherein the nucleic acid molecules of (i) and (ii) are RNA molecules or DNA molecules.

17. The composition of embodiment 16, wherein the nucleic acid molecules of (i) and (ii) are ssDNA molecules or dsDNA molecules.

18. The composition of embodiment 16, wherein the nucleic acid molecules of (i) and (ii) are ssRNA molecules.

19. The composition of any one of embodiments 1-18, wherein the nucleic acid molecule of (ii) comprises two or more transgenes, and the transgenes are separated by one or more internal ribosome entry sites (IRESes) and/or one or more 2A peptide-encoding sequences.

20. The composition of any one of embodiments 1-19, wherein the nucleic acid molecule of (ii) further comprises one or more enhancers.

21. The composition of embodiment 20, wherein the one or more enhancers comprise a woodchuck hepatitis virus posttranscriptional regulatory element.

22. The composition of any one of embodiments 1-21, wherein the one or more nucleic acid molecules of (i) and/or (ii) are RNA molecules and comprise one or more modifications selected from modified ribonucleosides, a 5'-7mG cap structure, and a poly (rA) tail.

23. The composition of embodiment 22, wherein the modified ribonucleoside is pseudouridine or a derivative of pseudouridine.

24. The composition of any one of embodiments 1-23, wherein the nucleic acid molecule of (ii) is an RNA molecule and further comprises one or more modifications selected from a 5'-7mG cap structure and a poly (rA) tail.

25. The composition of any one of embodiments 1-24, wherein the Pol proteins and/or LTRs are based on Pol proteins and/or LTRs are from human immunodeficiency virus (HIV), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), human foamy virus (HFV), murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV).

26. The composition of any one of embodiments 1-25, wherein the composition is packaged in a non-viral delivery system.

27. The composition of embodiment 26, wherein the composition is packaged in lipid nanoparticles.

28. A composition comprising:

- (i) a first RNA molecule comprising: a 5' untranslated region (UTR), a nucleic acid sequence encoding a retroviral Pol polyprotein, and a 3' UTR; and

- (ii) a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR;

wherein the composition does not contain nucleic acid sequences that express proteins encoded by retroviral rev and env genes, and wherein the composition is packaged in a non-viral delivery system.

29. The composition of embodiment 28, wherein the RNA molecules of (i) and (ii) are ssRNA molecules.

30. The composition of any one of embodiments 28-29, further comprising a third RNA molecule encoding one or more accessory proteins selected from nucleocapsid (NC), capsid protein (CA), viral infectivity factor (Vif), transactivator-of transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu).

31. The composition of embodiment 28, wherein the first RNA molecule further comprises a nucleic acid sequence encoding one or more accessory proteins selected from NC, CA, Vif, Tat, Nef, Vpr, and Vpu.

32. The composition of any one of embodiments 28-31, wherein the second RNA molecule comprises two or more transgenes, and the transgenes are separated by one or more internal ribosome entry sites (IRESes) and/or one or more sequences encoding a 2A peptide-encoding sequence.

33. The composition of any one of embodiments 28-32, wherein the second RNA molecule further comprises one or more enhancers.

34. The composition of embodiment 33, wherein the one or more enhancers comprise a woodchuck hepatitis virus posttranscriptional regulatory element.

35. The composition of any one of embodiments 28-34, wherein the first RNA molecule comprises one or more modifications selected from modified nucleosides, a 5'-7mG cap structure, and a poly (rA) tail.

36. The composition of any one of embodiments 28-35, wherein the second RNA molecule comprises one or more modification selected from a 5'-7mG cap structure and a poly (rA) tail.

37. The composition of any one of embodiments 1-36, wherein the one or more promoters comprise one or more tissue-specific or cell-specific promoters.

38. The composition of embodiment 37, wherein the one or more tissue-specific or cell-specific promoters are specific to bone marrow, hematopoietic stem cells (HSCs), epithelial cells, liver cells, ocular cells, muscle cells, or T cells.

39. The composition of any one of embodiments 1-36, wherein the one or more promoters comprise an hCMV promoter.

40. The composition of any one of embodiments 1-39, wherein the one or more transgenes encode one or more therapeutic molecules, diagnostic molecules, or reporter molecules, or fragments thereof.

41. The composition of embodiment 40, wherein the one or more transgenes encode one or more therapeutic proteins, diagnostic proteins, or reporter proteins, or fragments thereof.

42. The composition of embodiment 41, wherein the therapeutic protein is beta-globin, cystic fibrosis transmembrane conductance regulator (CFTR), Factor VIII, dystrophin, or RP GTPase regulator (RPGR).

43. The composition of embodiment 41, wherein the reporter protein is a fluorescent protein or luciferase.

44. The composition of any one of embodiments 38-43, wherein the Pol proteins, accessory proteins, and/or LTRs are based on Pol proteins, accessory proteins, and/or LTRs

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from human immunodeficiency virus (HIV), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), human foamy virus (HFV), murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV).

45. The composition of any one of embodiment 26 or 28-44, wherein the non-viral delivery system is targeted to a specific tissue or cell type.

46. The composition of embodiment 45, wherein the specific tissue or cell type is bone marrow, HSCs, epithelial cells, liver cells, ocular cells, muscle cells, or T cells.

47. The composition of any one of embodiments 28-46, wherein the non-viral delivery system is a lipid nanoparticle, a liposome, a polypeptide nanoparticle, a silica nanoparticle, a gold nanoparticle, a polymeric nanoparticle, a dendrimer, or a cationic nanoemulsion.

48. A method for expressing a gene in a subject in need thereof, the method comprising administering to the subject an effective amount of the composition of any one of embodiments 1-47, thereby expressing the one or more transgenes in the subject.

49. A method for expressing a gene in a cell, the method comprising delivering the composition of any one of embodiments 1-47 to the cell, thereby expressing the one or more transgenes in the cell.

50. A method of using the composition of any one of embodiments 1-47, the method comprising delivering the composition to a subject, thereby expressing the one or more transgenes in the subject.

51. A method of treating a disease or condition in a subject in need thereof, the method comprising delivering the composition of any one of embodiments 1-47 to the subject, thereby expressing the one or more transgenes in the subject.

52. The method of embodiment 51, wherein the disease or condition is a genetic disease or condition.

53. The method of embodiment 51, wherein the disease or condition is a hereditary genetic disease or condition.

54. The method of embodiment 51, wherein the disease or condition is sickle cell disease, beta-thalassemia, haemophilia B, retinitis pigmentosa, Duchenne muscular dystrophy, cystic fibrosis, or cancer.

55. The method of any one of embodiments 48-54, wherein the one or more transgenes are integrated into the genome of a target cell.

56. The method of embodiment 55, wherein the one or more transgenes are stably expressed for at least a week, at least two weeks, at least a month, at least 6 months, at least a year, or for the lifetime of the subject.

57. A method of eliciting an immune response in a subject in need thereof, the method comprising administering to the subject an effective amount of the composition of any one of embodiments 1-47, thereby expressing the one or more transgenes in the subject.

58. The method of embodiment 57, wherein the subject has cancer, and the one or more transgenes encode a tumor antigen.

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59. The method of embodiment 57, wherein the subject has or is at risk of contracting an infectious disease, and the one or more transgenes encode an antigen associated with the infectious disease.

60. The method of any one of embodiments 48-59, wherein the composition is delivered locally or systemically.

61. The method of embodiment 60, wherein the composition is delivered by injection, by inhalation, intravenously, intraperitoneally, subcutaneously, intramuscularly, orally, intranasally, by pulmonary administration, transdermally, transmucosally, or intratumorally.

62. One or more nucleic acid expression cassettes comprising a 5' UTR, a nucleic acid sequence encoding one or more retroviral Pol polyprotein components, and a 3' UTR, wherein the expression of Pol polyprotein components do not require translational slippage from an inline gag gene for use in the composition of any one of embodiments 1-47.

63. The nucleic acid expression cassette of embodiment 62, further comprising a nucleic acid sequence encoding one or more accessory proteins selected from nucleocapsid (NC), capsid protein (CA), viral infectivity factor (Vif), transactivator-of transcription (Tat), negative regulatory factor (Nef), viral protein R (Vpr), and viral protein u (Vpu).

64. The nucleic acid expression cassette of embodiment 62 or 63, wherein the Pol proteins and/or accessory proteins are based on Pol proteins and accessory proteins from: human immunodeficiency virus (HIV), simian immunodeficiency virus (SIV), visna/maedi virus (VMV), caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), feline immunodeficiency virus (FIV), bovine immunodeficiency virus (BIV), human foamy virus (HFV), murine leukemia virus (MLV), Moloney murine leukemia virus (MoLV), Friend virus (FV), Abelson murine leukemia virus (A-MLV), murine stem cell virus (MSCV), mouse mammary tumor virus (MMTV), Moloney murine sarcoma virus (MoMSV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), FBR murine osteosarcoma virus (FBR MSV), Avian myelocytomatosis virus-29 (MC29), Avian erythroblastosis virus (AEV), human T-cell leukemia virus (HTLV).

65. A method of producing an RNA molecule, comprising in vitro transcribing the nucleic acid expression cassette of any one of embodiments 62-64.

66. A kit comprising one or more containers, the one or more containers comprising the composition of any one of embodiments 1-47, or the nucleic acid expression cassette of any one of embodiments 62-65.

EQUIVALENTS AND SCOPE

While several inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the

foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

All references, patents, and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03. It should be appreciated that embodiments described in this document using an open-ended transitional phrase (e.g., “comprising”) are also contemplated, in alternative embodiments, as “consisting of” and “consisting essentially of” the feature described by the open-ended transitional phrase. For example, if the disclosure describes “a composition comprising A and B,” the disclosure also contemplates the alternative embodiments “a composition consisting of A and B” and “a composition consisting essentially of A and B.”

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 <213> ORGANISM: Artificial Sequence
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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

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<210> SEQ ID NO 7

<211> LENGTH: 1700

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 7

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tcggttaagg ccagggggaa agaaaaata taaattaaaa catatagtat gggcaagcag	180
ggagctagaa cgattcgcag ttaatcctgg cctgttagaa acatcagaag gctgtagaca	240
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taatacagta gcaacctctc attgtgtgca tcaaaggata gagataaaaag acaccaaggga	360
agctttagac aagatagagg aagagcaaaa caaaagtaag aaaaaagcac agcaagcagc	420
agctgacaca ggacacagca atcaggtcag ccaaaattac cctatagtgc agaacatcca	480
ggggcaaatg gtacatcagg ccatatcacc tagaacttta aatgcatggg taaaagtagt	540
agaagagaag gctttcagcc cagaagtgat acccatgttt tcagcattat cagaaggagc	600

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cacccccaaa gatttaaaca ccatgctaaa cacagtgggg ggacatcaag cagccatgca	660
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tgcagggcct attgcaccag gccagatgag agaaccaagg ggaagtgaca tagcaggaac	780
tactagtacc cttcaggaac aaataggatg gatgacacat aatccaccta tcccagtagg	840
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acaataaagc tcgctttctt gctgtccaat ttctattaaa gggtccttg ttcctaagt	1620
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<210> SEQ ID NO 8

<211> LENGTH: 1619

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 8

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agcaagctgc gataagtgtc aactgaaggg tgaagcgatg cacggtcaag tggactgttc	240
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agaattcgga atcccgata acccgagag tcaggggcgtt attgagagca tgaacaagga	540
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aatggcggtg ttcatacata attttaagcg gaaggaggga attggaggat actctgcggg	660
agagaggata gtggatataa tagcgaccga tattcagaca aaggagctgc aaaaacagat	720
aacgaaaata caaaatttct gagtttacta tcgggactcc cgcgatcccc tgtggaaagg	780
tccagcgaaa ttgctttgga agggcggaag cgcggtagt atccaggaca attctgatat	840
caaagtggtc ccaaggcgga aagcaaagat aatccgcgac tacggcaagc aaatggcagg	900
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actaaactgg gggatattat gaaggccctt gagcatctgg attctgccta ataagaaaca	1560
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<210> SEQ ID NO 9

<211> LENGTH: 3764

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 9

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ttcagagcag accagagcca acagcccccac cagaagagag cttcaggttt ggggaagaga	180
caacaactcc ctctcagaag caggagccga tagacaagga actgtatcct ttagcttccc	240
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cgtcaacatc atcggagcca acctgttgac gcagatcggt tgcaagctga acttcccacat	540
tagccctatc gagacggtac cggatgaagct gaagcccgga atggacggcc cgaaggtcaa	600
gcaatggcca ttgacagagg agaagatcaa ggcactggtg gagatttgca cagagatgga	660
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<210> SEQ ID NO 10
 <211> LENGTH: 4474
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 10

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<210> SEQ ID NO 11
 <211> LENGTH: 3260
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 11

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 gcgcacgcaa gacttctggg aggttcagct gggcatcccg caccocgag ggctgaagaa 840
 gaagaaatcc gtgaccgtac tggatgtggg tgatgcctac ttctccgttc ccttggacga 900
 agacttcagg aagtacactg ccttcacaat ccttcgata aacaacgaga caccggggat 960
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 gtacatggat gacttgtacg tgggctctga tctagagata gggcagcacc gcaccaagat 1140
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 ggatggggcc gctaacaggg agactaagct gggcaaagcc ggatacgtca ctaaccgggg 1920
 cagacagaag gttgtcacc tcactgacac caccaaccag aagactgagc tgcaggccat 1980

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ttacctcgct ttgcaggact cgggectgga ggtgaacatc gtgacagact ctgagtatgc	2040
cctgggcato attcaagccc agccagacca gagtgagtcg gagctggta atcagatcat	2100
cgagcagctg atcaagaagg aaaaggtcta tctggcctgg gtacccgccc aaaaaggcat	2160
tggcggcaat gagcaggctg acaagctggt ctcgctggc atcaggaagg tgctattcct	2220
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cagtgtctacg gttaaggccg cctgctggtg ggcggaatc aagcaggagt tcgggatccc	2640
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cggccaggtc agagatcagg ctgagcatct caagaccgcg gtccaaatgg cggtattcat	2760
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ctttgttccc taagtccaac tactaaactg ggggatatta tgaagggcct tgagcatctg	3180
gattctgcct aataagaaac atttattgtc attgcagaga cgcggccgcg cgtctcaatg	3240
aagagcgtcg acgcatgcgg	3260

<210> SEQ ID NO 12

<211> LENGTH: 3212

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 12

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cgccatgttt tttagggaa atctggcctt cccacaaggg aaggccaggg aattttcttc	120
agagcagacc agagccaaca gccccaccag aagagagctt caggtttggg gaagagacaa	180
caactccctc tcagaagcag gagccgatag acaaggaact gtatccttta gcttccctca	240
gatcactctt tggcagcgac cctcgtcac aataaagata ggggggcaat taaaggaagc	300
tctattagat acaggagcag atgatacagt attagaagaa atgaatttgc caggaagatg	360
gaaacccaaa atgatagggg gaattggagg ttttatcaaa gtaagacagt atgatcagat	420
actcatagaa atctgcggac ataaagctat aggtacagta ttagtaggac ctacacctgt	480
caacataatt ggaagaaatc tgttgactca gattggctgc actttaaat ttccttag	540
tcctattgag actgtaccag taaaattaaa gccaggaatg gatggcccaa aggttaacaa	600
atggccattg acagaagaaa aaataaaagc attagtagaa atttgtacag aaatggaaaa	660
ggaaggaaaa atttcaaaaa tggggcctga aaatccatac aatactccag tatttgccat	720
aaagaaaaaa gacagtacta aatggagaaa attagtagat ttcagagaac ttaataagag	780

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aactcaagat	ttctgggaag	ttcaattagg	aataccacat	cctgcagggt	taaaacagaa	840
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cttcaggaag	tatactgcat	ttaccatacc	tagtataaac	aatgagacac	cagggattag	960
atatcagtac	aatgtgcttc	cacagggatg	gaaaggatca	ccagcaatat	tccagtgtag	1020
catgacaaaa	atcttagagc	cttttagaaa	acaaaatcca	gacatagtca	tctatcaata	1080
catggatgat	ttgtatgtag	gatctgactt	agaaataggg	cagcatagaa	caaaaataga	1140
ggaactgaga	caacatctgt	tgagggtggg	atttaccaca	ccagacaaaa	aacatcagaa	1200
agaacctcca	ttcctttgga	tgggttatga	actccatcct	gataaatgga	cagtacagcc	1260
tatagtgtcg	ccagaaaagg	acagctggac	tgtcaatgac	atacagaaat	tagtgggaaa	1320
attgaattgg	gcaagtcaga	tttatgcagg	gattaaagta	aggcaattat	gtaaacttct	1380
taggggaacc	aaagcactaa	cagaagtagt	accactaaca	gaagaagcag	agctagaact	1440
ggcagaaaac	agggagattc	taaaagaacc	ggtacatgga	gtgtattatg	acctatcaaa	1500
agacttaata	gcagaaatac	agaagcaggg	gcaaggccaa	tggacatata	aaattttatca	1560
agagccattt	aaaaatctga	aaacaggaaa	gtatgcaaga	atgaagggtg	cccacactaa	1620
tgatgtgaaa	caattaacag	aggcagtaca	aaaaatagcc	acagaaagca	tagtaatatg	1680
gggaaagact	cctaaattta	aattacccat	acaaaaggaa	acatgggaag	catggtggac	1740
agagtattgg	caagccacct	ggattcctga	gtgggagttt	gtcaataccc	ctcccttagt	1800
gaagttatgg	taccagttag	agaagaacc	cataatagga	gcagaaactt	tctatgtaga	1860
tggggcagcc	aatagggaaa	ctaaattagg	aaaagcagga	tatgtaactg	acagagggaag	1920
acaaaaagtt	gtccccctaa	cggacacaa	aaatcagaag	actgagttac	aagcaattca	1980
tctagctttg	caggattcgg	gattagaagt	aaacatagtg	acagactcac	aatatgcatt	2040
gggaatcatt	caagcacaa	cagataagag	tgaatcagag	ttagtcagtc	aaataataga	2100
gcagttaata	aaaaaggaaa	aagtctacct	ggcatgggta	ccagcacaca	aaggaattgg	2160
aggaaatgaa	caagtagata	aattggtcag	tgctggaatc	aggaaagtac	tattttttaga	2220
tggaatagat	aaggcccaag	aagaacatga	gaaatatcac	agtaattgga	gagcaatggc	2280
tagtgatttt	aacctaccac	ctgtagtagc	aaaagaaata	gtagccagct	gtgataaatg	2340
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agattgtaca	catttagaag	gaaaagtatt	cttggtagca	gttcattgtag	ccagtggata	2460
tatagaagca	gaagtaattc	cagcagagac	agggaagaa	acagcatact	tcctcttaaa	2520
attagcagga	agatggccag	taaaaacagt	acatacagac	aatggcagca	atttcaccag	2580
tactacagtt	aaggccgctt	gttggtgggc	ggggatcaag	caggaatttg	gcattcccta	2640
caatccccaa	agtcaaggag	taatagaatc	tatgaataaa	gaattaaaga	aaattatagg	2700
acaggtaaga	gatcaggctg	aacatcttaa	gacagcagta	caaatggcag	tattcatcca	2760
caatttttaa	agaaaagggg	ggattggggg	gtacagtgca	ggggaaagaa	tagtagacat	2820
aatagcaaca	gacatacaaa	ctaaagaatt	acaaaaacaa	attacaaaaa	ttcaaaattt	2880
tcgggtttat	tacaggggaca	gcagagatcc	agtttgaaa	ggaccagcaa	agtcctctg	2940
gaaaggtgaa	ggggcagtag	taatacaaga	taatagtgc	ataaaagtag	tgccaagaag	3000
aaaagcaaag	atcatcaggg	attatggaaa	acagatggca	ggtgatgatt	gtgtggcaag	3060
tagacaggat	gaggattaaa	gctcgctttc	ttgctgtcca	atttctatta	aaggttcctt	3120

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tgttccctaa gtccaactac taaactgggg gatattatga agggccttga gcatctggat	3180
tctgcctaata aagaacatt tattgtcatt gc	3212

<210> SEQ ID NO 13
 <211> LENGTH: 1115
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 13

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gaattggcgg gccatggcct ccgacttcaa cctgccaccc gtcgtcgcta aggagatcgt	180
tgctagctgc gacaagtgcc agctgaaagg cgaggctatg cacgggcagg ttgattgctc	240
tcccggcatc tggcagctcg actgtactca cctggagggc aaggtcatcc tggtcgccgt	300
gcacgtggcc tctgggtaca tcgaggtga ggtcatccct gcagagactg gccaggagac	360
tgccatttct ctgctgaaac tggccggccg ttggcctgtg aagacagtgc acacagataa	420
cggtcccaac ttcacctcca ccactgtgaa ggctgcctgc tgggtgggctg ggatcaagca	480
ggagttcggg atccctata acccagctc tcagggcgtg atcgaatcca tgaacaagga	540
gctgaagaag atcatcgcc aggttcggga ccaggcagag cacctgaaga ctgcagtgca	600
catggccgtg ttcattcaca acttcaagcg aaagggcgcc atcggtgcct actcagccgg	660
cgagcggatc gtggacatca tcgccactga catccagacc aaagagctgc agaagcagat	720
caccaagatc cagaacttcc gtgtgtacta ccgggactcc cgggaccctg tgtggaaggg	780
ccctgccaa gctgtgtgga agggcgaggg cgccgtggtc attcaggaca actgtgacat	840
caaggttgtg cccaggcgca aggccaaag tatccgggac tacggcaagc agatggctgg	900
cgacgactgt gtggcctctc gtcaaatga ggactaaagc tcgctttctt gctgtccaat	960
ttctattaaa ggttctttg ttccctaagt ccaactacta aactggggga tattatgaag	1020
ggccttgagc atctggattc tgcctaataa gaaacattta ttgtcattgc agagacgcgg	1080
ccgcgcgtct caatgaagag cgtcgacgca tgcgg	1115

<210> SEQ ID NO 14
 <211> LENGTH: 1115
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 14

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cgccatggga ttcttgacg gcattgacaa ggccgaagag gagcacgaga aatatcatag	120
caactggagg gcgatggcat cagacttcaa cttgccgcca gtcgtggcta aagagatagt	180
agcaagctgc gataagtgtc aactgaaggg tgaagcagatg cacggtcaag tggactgttc	240
ccctgggtata tggcaactgg actgtacca tctcgaaggt aaagtcattt tggttgctgt	300
acatgttgcc tctgggtata tcgaggtga ggtgattcca gcagaaacgg gacaggaaac	360
agcttatttc ttgctcaagc ttgccgggag atggcctgtg aaaactgttc atacagataa	420
tggtagcaat tttactagta cgaccgtgaa agcagcttgt tgggtggcag ggattaagca	480
agaattcgga atcccgata acccgagag tcagggcgtt attgagagca tgaacaagga	540

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actgaaaaaa ataattgggc aagtgaagaga tcaggctgag catcttaaaa ctgctgtaca	600
aatggcggtg ttcatacata attttaagcg gaagggagga attggaggat actctgcggg	660
agagaggata gtggatataa tagcgaccga tttcagaca aaggagctgc aaaaacagat	720
aacgaaaata caaaattttc gagtttacta tcgggactcc cgcgatcccg tgtggaaagg	780
tccagcgaaa ttgcttttga agggcggaagg cgcggtagt atccaggaca attctgatat	840
caaagtggtc ccaaggcgga aagcaaagat aatccgcgac tacggcaagc aaatggcagg	900
agatgattgc gtcgcatcac gacaggacga agattaaagc tcgctttctt gctgtccaat	960
ttctattaaa ggttcttttg ttccctaagt ccaactacta aactggggga tattatgaag	1020
ggccttgagc atctggattc tgccataaa gaaacattta ttgtcattgc agagacgcgg	1080
ccgcgcgtct caatgaagag cgtcgacgca tgcgg	1115

<210> SEQ ID NO 15
 <211> LENGTH: 1067
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 15

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cgccatgttt ttagatggaa tagataaggc ccaagaagaa catgagaaat atcacagtaa	120
ttggagagca atggctagt attttaacct accacctgta gtagcaaaag aaatagtagc	180
cagctgtgat aaatgtcagc taaaagggga agccatgcat ggacaagtag actgtagccc	240
aggaatatgg cagctagatt gtacacattt agaaggaaaa gttatcttgg tagcagttca	300
tgtagccagt ggatatatag aagcagaagt aattccagca gagacagggc aagaaacagc	360
atacttcttc ttaaaattag caggaagatg gccagtaaaa acagtacata cagacaatgg	420
cagcaatttc accagtacta cagttaaggc cgctgttgg tggcgggga tcaagcagga	480
atttggcatt cctacaatc cccaaagtca aggagtaata gaatttatga ataaagaatt	540
aaagaaaatt ataggacagg taagagatca ggctgaacat cttagacag cagtacaaat	600
ggcagtattc atccacaatt taaaagaaa aggggggatt ggggggtaca gtgcagggga	660
aagaatagta gacataatag caacagacat acaaactaaa gaattacaaa aacaaattac	720
aaaaattcaa aattttcggg ttattacag ggacagcaga gatccagttt ggaaaggacc	780
agcaaagtc ctctggaaag gtgaaggggc agtagtaata caagataata gtgacataaa	840
agtagtgcca agaagaaaag caaagatcat cagggtattat ggaaaacaga tggcagggtga	900
tgattgtgtg gcaagtagac aggatgagga ttaaagctcg ctttcttgc gtccaatttc	960
tattaaaggt tcctttgttc cctaagtcca actactaac tgggggatat tatgaagggc	1020
cttgagcatc tggattctgc ctaataagaa acatttattg tcattgc	1067

<210> SEQ ID NO 16
 <211> LENGTH: 1568
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 16

agaccaggaa ttacatttgc ttctgacaca actgtgttca ctagcaacct caaacagagc	60
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cgccatgcc attagcccta tcgagacggg accggtgaag ctgaagcccg ggatggacgg	120
cccgaaggtc aagcaatggc cattgacaga ggagaagatc aaggcactgg tggagatttg	180
cacagagatg gaaaaggaag ggaaaatctc caagattggg cctgagaacc cgtacaacac	240
gccggtgttc gcaatcaaga agaaggactc gacgaaatgg cgcaagctgg tggacttccg	300
cgagctgaac aagcgcacgc aagacttctg ggaggttcag ctgggcatcc cgcaccccg	360
agggctgaag aagaagaaat ccgtgaccgt actggatgtg ggtgatgctt acttctccgt	420
tcccctggac gaagacttca ggaagtacac tgccttcaca atcccttcga tcaacaacga	480
gacaccgggg attcgatata agtacaacgt gctgcccag ggctggaaa gctctcccg	540
aatcttcag agtagcatga ccaaatcct ggagccttc cgcaaacaga accccgacat	600
cgatcatctat cagtacatgg atgacttgta cgtgggctct gatctagaga tagggcagca	660
ccgcaccaag atcaggagac tgcgccagca cctgttgagg tggggactga ccacaccga	720
caagaagcac cagaaggagc ctcccttctt ctggatgggt tacgagctgc accctgacaa	780
atggaccgtg cagcctatcg tctgtccaga gaaagacagc tggactgtca acgacataca	840
gaagctggtg gggaggtga actgggccag tcagatttac ccagggatta aggtgaggca	900
gctgtgcaaa ctctcccg gaaccaaggc actcacagag gtgatcccc taaccgagga	960
ggccgagctc gaactggcag aaaaccgaga gatcctaaag gagcccgctgc acggcgtgta	1020
ctatgacccc tccaggacc tgatgccga gatccagaag caggggcaag gccagtgga	1080
ctatcagatt taccaggagc cctcaagaa cctgaagacc ggcaagtacg cccggatgag	1140
gggtgccac actaacgacg tcaagcagct gaccgaggcc gtgcagaaga tcaccaccga	1200
aagcatcgtg atctggggaa agactcctaa gttcaagctg cccatccaga aggaaacctg	1260
ggaaacctgg tggacagagt attggcaggc cacctggatt cctgagtggg agttcgtcaa	1320
caccctccc ctggtgaagc tgtggtacca gctggagaag gagcccatag tgggcgccga	1380
aaccttctaa agctcgcttt ctgctgtcc aatttctatt aaagggttctt ttgttcctta	1440
agtccaacta ctaaaactgg ggatattatg aagggccttg agcatctgga ttctgcctaa	1500
taagaaacat ttattgtcat tgcagagacg cggccgcgcg tctcaatgaa gagcgtcgac	1560
gcatgcgg	1568

<210> SEQ ID NO 17

<211> LENGTH: 1928

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 17

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cccgaaggtc aagcaatggc cattgacaga ggagaagatc aaggcactgg tggagatttg	180
cacagagatg gaaaaggaag ggaaaatctc caagattggg cctgagaacc cgtacaacac	240
gccggtgttc gcaatcaaga agaaggactc gacgaaatgg cgcaagctgg tggacttccg	300
cgagctgaac aagcgcacgc aagacttctg ggaggttcag ctgggcatcc cgcaccccg	360
agggctgaag aagaagaaat ccgtgaccgt actggatgtg ggtgatgctt acttctccgt	420
tcccctggac gaagacttca ggaagtacac tgccttcaca atcccttcga tcaacaacga	480
gacaccgggg attcgatata agtacaacgt gctgcccag ggctggaaa gctctcccg	540

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aatcttcag agtagcatga ccaaaatcct ggagccttcc cgaaacaga accccgacat	600
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ccgcaccaag atcgaggagc tgcgccagca cctgttgagg tggggactga ccacaccga	720
caagaagcac cagaaggagc ctcccttccct ctggatgggt tacgagctgc accctgacaa	780
atggaccgtg cagcctatcg tctgtccaga gaaagacagc tggactgtca acgacatata	840
gaagctgggtg gggaaagtga actgggccag tcagatttac ccagggatta aggtgaggca	900
gctgtgcaaa ctctccgcg gaaccaaggc actcacagag gtgatcccc taaccgagga	960
ggccgagctc gaactggcag aaaaccgaga gatcctaaag gagcccgctgc acggcgtgta	1020
ctatgacccc tccaaggacc tgatcgccga gatccagaag caggggcaag gccagtggac	1080
ctatcagatt taccaggagc ccttcaagaa cctgaagacc ggcaagtacg cccggatgag	1140
gggtgcccac actaacgacg tcaagcagct gaccgaggcc gtgcagaaga tcaccaccga	1200
aagcatcgtg atctggggaa agactcctaa gttcaagctg cccatccaga aggaaacctg	1260
ggaaacctgg tggacagagt attggcaggc cacctggatt cctgagtggg agttcgtcaa	1320
caccctctcc ctggtgaagc tgtggtacca gctggagaag gagcccatag tgggcgccga	1380
aaccttctac gtggatgggg ccgctaacag ggagactaag ctgggcaaaag ccggatacgt	1440
cactaaccgg ggcagacaga aggttggtcac cctcactgac accaccaacc agaagactga	1500
gctgcaggcc atttacctg ctttgacagga ctccggcctg gaggtgaaca tcgtgacaga	1560
ctctcagtat gccctgggca tcattcaagc ccagccagac cagagtgagt ccgagctggt	1620
caatcagatc atcgagcagc tgatcaagaa ggaaaaggct tatctggcct gggatcccgc	1680
ccacaaaggc attggcgcca atgagcaggt cgacaagctg gtctcggtg gcacaggaa	1740
gggtgctataa agctcgcttt cttgctgtcc aatttctatt aaagggttcc ttgttcccta	1800
agtccaacta ctaaactggg ggatattatg aagggccttg agcatctgga ttctgcctaa	1860
taagaaacat ttattgtcat tgcagagacg cggccgcgcg tctcaatgaa gagcgtcgac	1920
gcatgcgg	1928

<210> SEQ ID NO 18
 <211> LENGTH: 18
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 18

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<210> SEQ ID NO 19
 <211> LENGTH: 18
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

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What is claimed is:

1. A composition comprising:

- (i) one or more nucleic acid molecules encoding two or more Pol polyprotein components flanked by 5' and 3' untranslated regions (UTRs), wherein the two or more Pol polyprotein components encoded by the one or more nucleic acid molecules comprise a reverse transcriptase and an integrase; and
- (ii) a ribonucleic acid (RNA) molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR,

wherein the composition does not contain nucleic acid sequences that express proteins encoded by either the retroviral rev gene or the retroviral env gene, or both.

2. The composition of claim 1, wherein the composition is capable of integrating the one or more transgenes into a host genome in the absence of functional retroviral Rev and/or Env proteins.

3. The composition of claim 1, wherein the expression of the one or more Pol polyprotein components does not require translational slippage from an inline gag gene.

4. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) is one nucleic acid molecule comprising a 5' UTR, a nucleic acid sequence encoding at least the Pol polyprotein components reverse transcriptase and integrase, and a 3' UTR.

5. The composition of claim 1, wherein the Pol polyprotein components reverse transcriptase and integrase are expressed on one nucleic acid molecule with one or more polycistronic elements.

6. The composition of claim 5, wherein the one or more polycistronic elements comprise an intervening internal ribosome entry site (IRES), a 2A peptide-encoding sequence, or both.

7. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) comprise nucleic acid sequences encoding one or more accessory proteins.

8. The composition of claim 1, wherein the composition does not comprise a nucleic acid sequence encoding the matrix protein.

9. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) comprise at least one mutation in one or more intrinsic instability (INS) elements.

10. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) encode an integrase polypeptide comprising an N-terminal methionine-glycine dipeptide.

11. The composition of claim 1, wherein the one or more Pol polyprotein components are fused to a homing protein.

12. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) and/or the RNA molecule of (ii) are codon optimized for expression in a host cell.

13. The composition of claim 1, wherein the composition further comprises a priming oligonucleotide.

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14. The composition of claim 1, wherein the composition is packaged in a non-viral delivery system.

15. The composition of claim 1, wherein the one or more promoters comprise one or more tissue-specific or cell-specific promoters.

16. A method for expressing a gene in a subject, the method comprising administering to the subject an effective amount of the composition of claim 1, thereby expressing the one or more transgenes in the subject.

17. A method for expressing a gene in a cell, the method comprising delivering an effective amount of the composition of claim 1 to the cell, thereby expressing the one or more transgenes in the cell.

18. A method of eliciting an immune response in a subject, the method comprising administering to the subject an effective amount of the composition of claim 1, thereby expressing the one or more transgenes in the subject that are capable of eliciting an immune response.

19. A kit comprising one or more containers, wherein the one or more containers comprises the composition of claim 1.

20. The composition of claim 1, wherein the one or more nucleic acid molecules of (i) is one or more RNA molecules.

21. The composition of claim 20, wherein the one or more RNA molecules comprise one or more modifications selected from modified ribonucleosides, a 5'-7 mG cap structure, and a poly (rA) tail; and/or the RNA molecule of (ii) further comprises one or more modifications selected from modified ribonucleosides, a 5'-7 mG cap structure, and a poly (rA) tail.

22. A composition comprising:

- (i) a first RNA molecule comprising: a 5' untranslated region (UTR), a nucleic acid sequence encoding two or more Pol polyprotein components, and a 3' UTR, wherein the two or more Pol polyprotein components comprise a reverse transcriptase and an integrase; and
- (ii) a second RNA molecule comprising one or more reverse transcriptase priming elements and one or more promoter sequences operably linked to one or more transgenes, between a 5' long terminal repeat (LTR) and a 3' LTR;

wherein the composition is packaged in a non-viral delivery system.

23. One or more nucleic acid templates comprising a 5' UTR, a nucleic acid sequence encoding two or more retroviral Pol polyprotein components, and a 3' UTR, wherein the two or more retroviral Pol polyprotein components encoded by the one or more nucleic acid templates comprise a reverse transcriptase and an integrase, and wherein the expression of Pol polyprotein components does not require translational slippage from an inline gag-pol gene.

24. A method of producing an RNA molecule, the method comprising transcribing in vitro the nucleic acid template of claim 23.

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